A field-level dark-siren likelihood

Making the large-scale-structure field a latent variable of the hierarchy, and recovering catalog sirens and cross-correlation sirens as its two limits

2026 August 10

Executive summary

The dark-siren likelihood in this repository already contains a three-dimensional Bayesian reconstruction of the galaxy density field. It is used in the wrong place. The field is fitted once, offline, at a fixed cosmology, a fixed mean density and a fixed completeness fit; it is exported to disk as a table $Q_p(z)$; and a provenance firewall then *forbids* the inference from sampling any of the parameters the table was built at. The literal sentence "every cosmological proposal changes the galaxy-field likelihood consistently" is, today, unrepresentable by construction.

The proposal. Delete the table. Promote the whitened low-rank coefficients ξ of the field to a latent variable of the hierarchy, so that one field appears in three places at once: in the galaxy *counts*, which constrain it; in the *placement* of missing hosts, which the gravitational-wave events probe; and in the *selection normalisation*, which decides what fraction of universes like this one are detectable. Every cosmological proposal then re-weights the galaxy-field likelihood, and the missing-host field it delivers to the event terms and to the detected fraction is the same realisation.

Why this is more than plumbing. The same generative hierarchy contains the two established dark-siren analyses as limits of one likelihood. When the catalog is complete the missing branch vanishes and the field drops out *identically*: one recovers the discrete weighted-host sum. When the catalogued fraction of an event's localisation volume tends to zero and the volume spans many correlation cells, marginalising the Gaussian latent analytically replaces the field by its kernel, and the surviving event $\times$ galaxy-voxel term is the harmonic cross-correlation $C_\ell^{\mathrm{GW}\times\mathrm{g}}$ carrying the bias combination $b_{\mathrm{GW}} b_{\mathrm{gal}}$. The interior — an incomplete catalog with well-localised events — is where every real catalog/detector pairing actually sits, and it is the regime neither limit describes. Which corner a given pairing occupies is set by two numbers one can measure, not by which method one prefers.

What is affordable, and what is not. A full per-proposal Newton re-solve of the field is not affordable — though not for the reason first written down. The production baseline is now measured on the line this work targets: 1417 ms per likelihood call with no structure at all, on one H100 (§3), not the 27.5–49.3 ms that belongs to a different configuration. A ~ 1 s re-solve at the guard-permitted rank is therefore of order the likelihood itself, not 36 times it, and the refusal has to rest — and does rest — on memory and on smoothness rather than on wall clock. The affordable form is *linear response*: the mode of the field posterior moves affinely with the proposal through $S = \partial\hat{\xi}/\partial\theta$, obtained from the implicit function theorem against the Cholesky factor the anchor solve already computes, at a measured storage cost of 101 kB in the shipped artefact (five columns at $M = 2520$) and an online cost of one member-independent row-factor contraction. The correction is member-independent, which is the single fact that keeps the memory budget flat under a 256-fold evaluation concurrency.

What must be conceded in the same breath. Three things, and they belong in the abstract, not the appendix. (i) The galaxy field constrains the *shape* parameters $(\Omega_m, w_0, w_a, \delta)$ and the tracer biases; it constrains H_0 only through the volume–density amplitude $n_0(c/H_0)^3$, which is exactly degenerate with the mean comoving number density — so an H_0 constraint from the counts is manufactured unless n_0 carries an H_0-aware prior. (ii) The field is defined over the volume the survey sees: $Q \equiv 1$ off the footprint (38% of the sky) and above the survey depth $z_{\rm depth} = 0.3$. Four independent integrals put the in-support fraction of the missing-galaxy budget at 2.2×10^{-5} to 2.7×10^{-4}. The lowest and most defensible is the verification integral for this document, which uses the shipped fit *including its cubic $K(z)$ template* on the production $z_{\rm max} = 6$ grid: 2.2×10^{-5} all-sky, 1.4×10^{-5} restricted to the occupied footprint. The K-correction is why it sits below the other three: it drives $C_{\rm sel}$ down faster with redshift ($C_{\rm sel} = 0.92$ at $z_{\rm depth}$, $< 10^{-3}$ by $z = 1$), moving yet more of the missing budget above the survey depth, where $Q \equiv 1$. The field is being asked to redistribute two thousandths of one percent of the missing budget — but that is the *budget* share, and it is the wrong statistic for whether the field can move the answer, because the events sit exactly where the catalog is. Measured, 24.6% of all posterior samples fall in support and the event-weighted missing-branch mass is $\sum_i \varphi_i = 0.993$, two to three orders above the budget share (§3). (iii) The mean-one budget constraint that makes "C and n_0 own the budget, Q owns placement" true is a *gauge fixing*, not a measurement.

The cheapest thing to do was done first, and it did not stop the programme. Point (ii) was written not as a caveat but as a falsifiable prediction that the whole direction might be a null operation, testable in three days against artefacts that existed already: the event-weighted fraction of prior mass in the in-support missing branch, and the oscillation of the log-likelihood across the H_0 prior when the *existing* table is switched on at the anchor. Both were measured and the prediction is refuted. The event-weighted mass is $\sum_i \varphi_i = 0.993$ against a kill threshold of 10^{-3}, and the oscillation is 10.85 nats against a threshold of 0.1: both legs miss by two orders of magnitude. So the upper bound on everything the entire programme can buy is 10.85 nats rather than zero — 0.3 nat and $0.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ of median shift within the posterior bulk. Every other measurement in this document is downstream of that one, and §3 reports them.

What came back, successes and failures in the same breath. Rungs 0 through 8 have run (§3). The field's θ-dependence on the gravitational-wave channel is 5.4×10^{-4} nat, 300 times under its own gate, so the θ-coupled rung is demoted to an inertness flag and the frozen-anchor ensemble is the deliverable; the galaxy-side evidence oscillates by 59.9 nats over the same draws and stays a diagnostic. The member marginalisation ships: the spread is 1.5×10^{-2} nat at the anchor and at most 1.15 nats anywhere in the H_0 prior, and the θ-varying bias at the shipped $M_{\rm draw} = 8$ is 7.1×10^{-3} nat against a 0.1-nat budget. The seam costs +2.7% of the measured baseline. Two things did not pass. The mock closure's coverage tiers fail — but they fail identically in the arm with *no* field, 82–92% of the scatter is the event draw at fixed catalog, and the fix aimed at them was implemented and measured not to help, so what those tiers measure is the mock's own event-channel calibration and not the latent design. And a 5% shared membership between two tracers breaks the bias-ratio recovery at 10σ while the quoted uncertainty *shrinks*. The honest one-line status is that the code is validated and the mock's parameter-estimation calibration is not.

What is genuinely new, stated at the level the literature permits. Three-dimensional Bayesian reconstruction of the galaxy field for dark sirens is not new. Lognormal density reconstruction is not new. Multiplicative large-scale-structure corrections to catalog completion are not new. What is not found in the completion literature is the conjunction: a *shared* latent field carrying per-tracer bias and per-tracer parametric selection, marginalised inside a finite-event gravitational-wave hierarchy with the same realisation in the event terms and in the detection normalisation, at fixed missing-budget monopole — and the demonstration that catalog sirens and cross-correlation sirens are its two limits. No unqualified claim to priority is made here, and none is made on assertion: the paper-by-paper search that licenses what *is* claimed was run as the opening item of work rather than the closing one, and is recorded in `pr0/NOVELTY_SEARCH.md` — ~ 22 papers individually assessed, search date 2026-08-13, each against the three tests (per-realisation budget constraint; realisations through the *selection* normalisation; matched multi-tracer realisations of one shared latent). Every statement below that could be read as priority carries that search behind it (§3, D11).

1 What the machinery does today

The pieces are usually described as three machines — a builder, a table, a likelihood. They are one generative model, and writing them as one is what exposes where the model is exact and where it is a plug-in. All equations marked [CODE] are transcribed from verified source lines at `master 0c5b3db`.

1.1 Notation

Sky pixels are $p \in \{1, \ldots, N_{\rm pix}\}$ in HEALPix RING ordering at $N_{\rm side} = 64$ ($N_{\rm pix} = 49\,152$, $\Omega_{\rm pix} = 4\pi/N_{\rm pix}$); redshift is carried on a grid of $N_{\rm grid} = 1086$ nodes; a voxel is $v = (p, z)$; coarse radial shells used by the field fit are $g \in \{1, \ldots, G_s\}$; catalogs (tracers) are $k \in \{1, \ldots, K\}$. Cosmological and population parameters are Λ (containing H_0); survey parameters are $\theta = (\log_{10} n_0, \delta, \theta_{\rm sel}, b_{\rm miss}, \ldots)$. The expected all-galaxy comoving density is

$$\bar{n}(z; \Lambda, \theta) \;=\; \frac{dN_{\rm exp}}{dz\, d\Omega} \;=\; n_0\, (1+z)^{\delta}\, \frac{dV_c}{dz\, d\Omega}(z; \Lambda), \tag{1}$$

and $C(z; \theta) \in [0, 1]$ is the completeness. The production configuration uses the parametric $C_{\rm sel}(z; \theta_{\rm sel})$ built from a truncated luminosity function; $f_p = 1 - \texttt{masked\_frac}_p \in [0, 1]$ is the per-pixel areal completeness, which is $f_p \equiv 1$ in today's code.

The production run this work targets carries 259 events at 4096 posterior samples, $1\,067\,946$ detected injections, 20 free dimensions, and *no large-scale structure at all*: it runs with the field switched off and no completion table. No production run on this line has ever carried structure of any kind; the field-level path would be the one that does, which is why the standard is "provably inert when off, provably correct when on."

1.2 Thinning: why the catalog and the missing galaxies separate

Conditional on a latent log-density contrast $s(p, z)$, galaxies are an inhomogeneous Poisson process with the lognormal intensity

$$\Lambda_{\rm gal}(p,z) \;=\; \bar n(z;\Lambda,\theta)\,\exp\Big[b_{\rm gal}\,s(p,z) - \tfrac{1}{2}b_{\rm gal}^2\sigma^2(p,z)\Big], \qquad \sigma^2(p,z) \equiv {\rm Var}[s(p,z)], \tag{2}$$

where the shift $-\frac{1}{2}b^2\sigma^2$ enforces $\mathbb{E}_{\rm prior}[e^{bs-\frac{1}{2}b^2\sigma^2}] = 1$: the field *redistributes* galaxies, it does not create them. This mean-one convention is the most load-bearing convention in the stack and it reappears three times below.

The survey observes each galaxy independently with probability $f_pC(z;\theta)$. By the Poisson thinning theorem the catalogued and missing galaxies are two *independent* Poisson processes,

$$\Lambda_{\rm cat}(p,z) = f_p\,C(z;\theta)\,\Lambda_{\rm gal}(p,z), \tag{3}$$
$$\Lambda_{\rm miss}(p,z) = \big[1 - f_p\,C(z;\theta)\big]\,\Lambda_{\rm gal}(p,z), \tag{4}$$

and, exactly,

$$p\big(\text{missing}\,\big|\,\text{catalogued},\,s\big) \;=\; p\big(\text{missing}\,\big|\,s\big). \tag{5}$$

Equation (5) says: once the field is fixed, the catalog carries no further information about what is missing. Everything the catalog knows about the missing galaxies, it knows through s. It is the entire resolution of the double-conditioning question of §2.4.

1.3 The redshift–sky prior as it is coded

The likelihood does not integrate (3); it uses the catalog itself. For a dark siren at (z, p),

$$p\big(z,p\,\big|\,\Lambda,\theta,Q\big) \;=\; \frac{N_{{\rm obs},p}\,\mathcal{K}_p(z) \;+\; \big[1 - C(z;\theta)\big]\,\bar n(z;\Lambda,\theta)\;Q_p(z)}{\mathcal{Z}(\Lambda,\theta,Q)}, \tag{6}$$

[CODE] with $N_{{\rm obs},p}$ the integer galaxy count in pixel p and $\mathcal{K}_p$ the pixel's photometric-redshift kernel density. The consumed missing density is $dN_{\rm miss} = [1-C]\,\bar n\,Q$, relaxed to $dN_{\rm exp}$ above the survey depth $z_{\rm depth}$ — i.e. $C := 0$, $Q := 1$ there. Comparing with (4) identifies

$$Q_p(z) \;\equiv\; \exp\Big[b_{\rm miss}\,s(p,z) - \tfrac{1}{2}b_{\rm miss}^2\sigma^2(p,z)\Big] \quad \text{with } f_p \equiv 1. \tag{7}$$

So Q *is* the exponentiated latent field, mean-one normalised, and nothing else. Its only defect as written is that it is a fixed table read from disk rather than a function of the sampled parameters.

Two structural consequences of $f_p \equiv 1$ deserve naming, because both are first-order for the target survey. The measured areal masking has mean 0.137, standard deviation 0.104 and 99th percentile 0.635 over the occupied footprint, while the magnitude depth is uniform at the retention cut ($m_{\rm th}$ percentiles 1, 50, 99 all equal 21.000; the imaging depth is nowhere the limit). So the entire per-pixel selection variation is *areal*, which is achromatic in redshift and magnitude and therefore exactly correctable. First, putting f_p on the count side only would invert the sign of the inferred incompleteness: a pixel with 63% areal masking would be told that 63% of its galaxies are a *density deficit* rather than a *selection deficit*, and that deficit re-enters the event likelihood as a claim about where hosts are not. Second, the dominant term is not the partial pixels: it is the 18 682 of 49 152 pixels with $f_p = 0$, whose missing-budget weight moves from $1 - C$ to 1, because a pixel with no catalog has *all* of its galaxies missing.

1.4 The offline field: Poisson–lognormal on a low-rank basis

Let $\{Z_j\}_{j=1}^{M}$ be inducing nodes on $\mathbb{S}^2 \times \zeta$, $\zeta = \log(1+z)$ — a Fibonacci sphere of $M_{\rm sph}$ points crossed with M_z nodes uniform in ζ — and $k = k_{\rm sph} \otimes k_\zeta$ a separable kernel. With $LL^\top = K_{ZZ} + j\nVdash$, the Nyström design and the whitened field are

$$\Phi = K_{XZ} L^{-\top} \in \mathbb{R}^{V\times M}, \qquad s = \Phi\xi, \qquad \xi \sim \mathcal{N}(0, \nVdash_M), \tag{8}$$

so $\mathrm{Cov}(s) = \Phi\Phi^\top$ and $\sigma^2_{\rm prior}(v) = \sum_i \Phi^2_{vi}$. Φ factorises exactly as $\Phi_{\rm sph} \otimes \Phi_\zeta$ and is never materialised: a dense Φ at $M = 3780$ would be 47.6 GB, and a build at $M = 1728$ has already exhausted memory at 21.7 GB.

The rank is not free. The angular and radial lengthscales must be equal in comoving units or the kernel is a pancake, and the radial scale must survive the photometric-redshift scatter. The measured scatter is $\sigma_z = 0.0227$ (median over 2.24M galaxies), i.e. $\sigma_\chi = 89\,$Mpc at $z_{\rm ref} = 0.237$, so a 50 Mpc radial kernel retains $\exp[-(89/50)^2/2] = 0.20$ of its own radial signal while a 190 Mpc kernel retains 0.90. The self-consistent configuration is isotropic at $\sim 190\,$Mpc: $\ell_{\rm ang} = 0.2$, $\ell_\zeta = 0.039$, $M_{\rm sph} = 315$, $M_z = 8$–12, $M = 2520$–3780; the shipped anchor is built at the lower end, $M = 315 \times 8 = 2520$. At that rank the measured spectrum of H on the real catalog puts 1497 of 2520 modes above eigenvalue 10 and only 199 below 1.01, at a per-mode amplitude $\|\hat{\xi}\|/\sqrt{M} = 2.46$: the count channel is strongly data-constrained, a few hundred modes are pure prior, and that split is what the Laplace covariance encodes.

The fit is over solve voxels $v = (p, g)$ with integer counts N_v and a base $\bar{\lambda}_v = \int_g C\,\bar{n}\,dz$, giving

$$J(\xi) = \tfrac{1}{2}\|\xi\|^2 + \sum_v \big[\lambda_v - N_v \log\lambda_v\big], \qquad \lambda_v = \bar{\lambda}_v \exp\Big[b(\Phi\xi)_v - \tfrac{1}{2} b^2 \sigma^2_{\rm prior}(v)\Big], \tag{9}$$

[CODE] with $\nabla J = \xi + b\Phi^\top(\lambda - N)$ and $H = \nVdash + b^2\Phi^\top \mathrm{diag}(\lambda)\Phi \succeq \nVdash$. The problem is convex and well posed even when $V < M$. The Laplace posterior $\mathcal{N}(\hat{\xi}, H^{-1})$ is sampled as

$$\xi_m = \hat{\xi} + L_H^{-\top} g_m, \qquad g_m \sim \mathcal{N}(0, \nVdash_M), \qquad L_H L_H^\top = H. \tag{10}$$

Lemma 1 (posterior-mean Q and its variance shift)**.** *Under (10), for the Nyström row ϕ_u at an output voxel u,*

$$\log \mathbb{E}[Q_u] = b\,\phi_u^\top \hat{\xi} - \tfrac{1}{2} b^2 \big[\sigma^2_{\rm prior}(u) - \sigma^2_{\rm post}(u)\big], \qquad \sigma^2_{\rm post}(u) = \|L_H^{-1}\phi_u\|^2. \tag{11}$$

Equation (11) *is* the table the likelihood consumes today. It has a physically desirable property that a MAP table does not: a voxel with no data has $\sigma_{\rm post} \to \sigma_{\rm prior}$ and $\phi_u^\top\hat{\xi} \to 0$, so $Q \to 1$. *No information, no modulation.*

Remark 1 (the plug-in gap, stated exactly). The event likelihood is a nonlinear functional of Q: Q appears inside a logarithm in (6) and inside $\log\mu$ in the detection normalisation. Hence $\log \mathbb{E}_\xi[\mathcal{L}(Q(\xi))] \neq \log \mathcal{L}(\mathbb{E}_\xi[Q])$, and running the deterministic table is a plug-in, exact only to first order in the posterior spread of s.

1.5 The mean-one budget constraint is a gauge fixing

Proposition 1 (Q has no monopole freedom)**.** *For any positive $\alpha(z)$, the map $Q_p(z) \mapsto \alpha(z) Q_p(z)$ together with $[1 - C]\bar{n} \mapsto [1 - C]\bar{n}/\alpha(z)$ leaves $dN_{\rm miss}$, and therefore the entire event likelihood, exactly invariant. The per-z monopole of Q is therefore not identified by the gravitational-wave data; it is identified only by whatever fixes $(n_0, \delta, \theta_{\rm sel})$.*

The code fixes this gauge at build time by rescaling each realisation so that

$$\frac{\sum_{p\in\mathcal{F}} w_p(z)\, Q_p(z)}{\sum_{p\in\mathcal{F}} w_p(z)} = 1, \qquad w_p(z) = \big[1 - C_p(z)\big]\, \bar{n}(z), \tag{12}$$

over the *fitted footprint* $\mathcal{F}$, applied to the deterministic table and to *each member independently*. This is necessary and it is measured: the Jensen contribution of Lemma 1 otherwise inflates the missing budget by +55%, because the posterior variance is largest exactly where the data are sparse.

Two precisions matter for what may be claimed. First, in the production completeness modes the constraint is *already* exactly the identity the likelihood consumes: the builder fits the whole sky with p-independent weights, so mean-one reduces to $\langle Q\rangle_{\rm sky} = 1$ and the consumed sum equals its $Q \equiv 1$ value for every z, member and θ. Under *stratified* selection the weights are p-dependent and evaluated at the build-time fit, so that identity holds only at $\theta = \theta_{\rm ref}$ and leaks elsewhere at the order of the stratum-to-stratum spread in $C_{\rm sel}$. Second, and more importantly for the paper: by Proposition 1 this is a *convention that makes an estimand well posed*, i.e. a definition, not a measurement. "C and n_0 own the budget, Q owns placement" is the correct sentence, and "gauge fixing" is the correct word for it.

Remark 2 (renormalisation does not commute with the ensemble mean). Writing Π for the renormalisation, Π is nonlinear in Q, so $\frac{1}{M}\sum_m \Pi[Q^{(m)}] \neq \Pi[\mathbb{E}[Q]]$ at order $\mathrm{Var}_{\rm sky}(Q)/N_{\rm pix}$ per shell. "Members carry placement uncertainty but zero budget uncertainty" is therefore a modelling choice, not a theorem.

1.6 Member marginalisation, and the price of a logarithm

With $M_{\rm draw}$ realisations the likelihood is

$$\log \mathcal{L}(\Lambda, \theta) = \operatorname*{LSE}_m \log \mathcal{L}_m - \log M_{\rm draw}, \tag{13}$$

[CODE] where the member index enters *both* the per-event terms and the detection normalisation μ — enforced as an all-or-none guard, and correct: a member is one realisation of the universe, and the same universe must place the hosts and set the detected fraction.

Proposition 2 (Jensen bias of (13))**.** *(13) is unbiased for $\mathcal{L}$ and biased for $\log\mathcal{L}$. If $\log\mathcal{L}_m$ has spread σ nats and is approximately Gaussian across members, then*

$$\mathbb{E}\big[\log \widehat{\mathcal{L}}_M\big] - \log\mathcal{L} \simeq -\frac{e^{\sigma^2}-1}{2M}, \qquad \frac{\mathrm{ESS}}{M} \simeq e^{-\sigma^2}, \qquad \Longrightarrow \qquad \boxed{M > \frac{e^{\sigma^2}-1}{2\epsilon}} \tag{14}$$

for an ϵ-nat bias.

This is sharper than the usual $M \sim e^{\sigma^2}$ folklore and it is brutal. At $\sigma = 1.0$ nats, $\epsilon = 0.1$ needs $M = 9$; at $\sigma = 1.5$, $M = 43$; at $\sigma = 2.6$, $M = 4.3\times 10^3$, and $M = 8$ carries a 54-nat bias. The member spread was therefore the largest open number in the design, and §2.10 predicts it in closed form and then measures it. The measurement excludes the $\sigma = 2.6$ branch outright: the worst node anywhere in the H_0 prior is 1.15 nats and the anchor is 1.5×10^{-2}. It also shows where (14)'s lognormal premise fails, and that it fails safely: at the low-H_0 rail the member log-weights are strongly *left*-skewed and heavy-tailed (skew −7.43, excess kurtosis 74.1), σ there is set by a handful of catastrophic members while the bulk of the ensemble is tight, and the measured $\mathrm{ESS}/M = 0.835$ against a predicted $e^{-\sigma^2} = 0.265$. On this line (14) is an upper bound, not an estimate.

Remark 3 (what actually has to be gated). The whitened draws g_m are generated once and reused at every θ (common random numbers), so $\widehat{\mathcal{L}}_M(\theta)$ is a *deterministic, smooth* function of θ, not a noisy one. The bias of Proposition 2 is a fixed perturbation and only its θ-variation, $\operatorname{osc}_\theta \Delta \equiv \max_\theta \Delta - \min_\theta \Delta$ with $\Delta(\theta) = \log \widehat{\mathcal{L}}_M - \log \mathcal{L}$, distorts the posterior; a constant offset is absorbed into the evidence. The consequence is unforgiving: every determinism diagnostic — repeat-to-repeat identity, adjacent-θ smoothness — passes *by construction*, and passes just as well on a badly distorted surrogate. Only $\operatorname{osc}_\theta \Delta$ against a large-M reference discriminates. This is not a hypothetical either: measured, 100 of 100 repeats are bit-identical in both the latent and the table arm, while the adjacent-θ smoothness statistic on a 10^4-point fixture sweep fails in *both* arms at the same point by the same factor, with the two curves agreeing to 5×10^{-11} — a statistic that cannot tell the two apart is not evidence about either.

1.7 The shared-latent multitracer solve, and which half of the seam is open

For K catalogs sharing one field the stacked objective and Hessian are

$$J(\xi) = \tfrac{1}{2}\|\xi\|^2 + \sum_k \sum_{v \in \mathcal{V}_k} \big[\lambda_{kv} - N_{kv} \log \lambda_{kv}\big], \qquad H = \nVdash + \sum_k b_k^2\, \Phi_k^\top \mathrm{diag}(\lambda_k) \Phi_k, \tag{15}$$

with $\lambda_{kv} = \bar{\lambda}_{kv} e^{b_k (\Phi_k \xi)_v - \frac{1}{2} b_k^2 \sigma_{kv}^2}$, one Laplace ensemble, and per-survey tables $\log Q_k^{(m)} = b_k \phi^\top \xi_m - \frac{1}{2} b_k^2 \sigma_{\mathrm{prior}}^2$. Member m of every catalog is then the same physical realisation — a theorem, not a check.

The interesting statement is where this stands operationally, and it is not where one would guess. **The consumer is ready; the producer cannot produce.** On the inference side, $K \geq 2$ mixtures under parametric selection completeness assemble and evaluate end to end: per-catalog completeness modes, per-catalog selection fits with an all-or-nothing anchoring rule, a homogeneous luminosity-function family, per-catalog parameter routing, and a two-catalog end-to-end likelihood fixture pinning the identical-catalogs-collapse-to-K=1 identity. Exactly one refusal survives on that side — *stratified* fits at $K \geq 2$ — and its stated reason is data plumbing, not modelling.

On the builder side the shared-field path is closed in four independent ways. It takes the per-pixel completeness default and is hard-rejected at load by any parametric-selection run; it never fixes the budget gauge, so every joint file ships the full +55% Jensen monopole that the single-catalog tables remove; its inducing grid is hard-coded at $M_{\mathrm{sph}}, M_z = 32, 6$ up to $z = 3$, giving a ζ node spacing of 0.277 and hence a *hard* floor of $L_{\mathrm{smooth}} \geq 1.34\,\mathrm{Gpc}$ — every physically supportable correlation length fails it; and it is the only producer of a shared realisation identifier, so K independently built single-catalog tables carry K different ones. The only way to run the configuration is a flag whose own documentation states that it marginalises over an *independent-fields product prior* rather than the shared-field prior the estimator assumes.

A shared-latent multitracer marginalisation under parametric selection completeness is not refused. It is not constructible.

This is the cleanest one-line motivation for the whole proposal: the field-level hierarchy closes that seam *by deleting the producer*. One ξ shared across K tracers makes matched realisations a theorem, retires the realisation identifier and the provenance firewall together, and removes the builder's frozen rank, frozen cosmology and frozen completeness mode simultaneously. That is now measured rather than argued: two disjoint tracers over one shared ξ recover an imposed bias ratio $b_2/b_1 = 2$ in 20 of 20 realisations at $\sigma_r = 0.0084$, against a deliberately decoupled two-field variant that returns 1.456 at $\sigma_r = 1.08$ — biased and uninformative at once (§3).

1.8 Novelty, scored honestly

The scoring below is the owner's, transcribed; the right-hand column is the correction this document's verification requires.

claim	score	reality check
homogeneous missing galaxies	none	—
$1 + b\delta_g$ correction	low	—
a multiplicative structure field Q	low	variance-completion work introduced it
clustering to reconstruct missing galaxies	not novel	—
Poisson/lognormal density reconstruction	not novel broadly	—
3-D Bayesian dark-siren catalog reconstruction	**not novel broadly**	cartography-style reconstructions already propagate voxel-count uncertainty and marginalise cosmology and bias; on the θ-coupling axis they are *ahead* of an exported table
low-rank sphere $\times\, z$ GP implementation	interesting implementation	—
posterior-mean rather than MAP Q	strong technical choice	correct, and it is what makes $Q \to 1$ where there is no data
per-z mean-one budget constraint	interesting methodological contribution	**it is a gauge fixing** (Prop. 1) — a definition, not a measurement; demote to an enabling clause
empty pixels as Poisson information	important, not fundamentally new	—
parametric selection C separated from Q	quite interesting combination	correct, and it is what makes the count likelihood non-circular
marginalising members through a complete GW hierarchy	**potentially novel**	the same-realisation requirement is a *correctness* condition, already enforced; the contribution is doing it at fixed budget monopole
joint multitracer shared latent	**potentially very novel**	**not constructible on the table path** (§1.7); constructible, and measured, on the latent path (§3)
joint multitracer + selection C + GW marginalisation	**strongest direction**	**implemented at mock scale** (§3); the field-level hierarchy was the shortest route to it, and a paper-by-paper search found no prior implementation

Table 1: Novelty scoring, with the corrections the code-level verification forces. Nothing in this document may be written as if three-dimensional Bayesian density reconstruction for dark sirens were new; that ground is taken, and the row in bold is where it is taken.

The claim this scoring is sharpest about is the budget-preserving one: posterior realisations $Q^{(m)}$ of the clustered missing field, each constrained to carry zero missing-budget monopole, marginalised through *both* the event terms and the detection normalisation. Two scope sentences must ride with that sentence wherever it is written — abstract included, not only here and not only in a risk table. First, the marginalisation is over *the volume the survey sees*, not over the budget: $Q \equiv 1$ with *zero* assumed variance off the footprint (38% of the sky) and above $z_{\rm depth} = 0.3$, and those two regions together carry $\sim 99.99\%$ of the missing-galaxy budget, so the statement "we marginalise over the clustered missing-galaxy field" is true over the volume the survey sees and false over the volume that carries the budget. Second, the compression that makes a defensible M_{draw} affordable is engineering, not a methods claim.

2 Proposed work

2.1 The hierarchy

Promote ξ from a table to a latent variable:

$$\begin{aligned} p\big(H_0, \Lambda, \xi, \{\theta_k\}, \{b_k\}, b_{\mathrm{GW}}, \{f_k\} \,\big|\, d_{\mathrm{GW}}, \{d_{\mathrm{gal},k}\}\big) \;\propto\; \pi(H_0,\Lambda)\,\pi(\{\theta_k\})\,\pi(\{b_k\})\,\pi(b_{\mathrm{GW}}) \\ \times\; \underbrace{\mathcal{N}(\xi; 0, \nVdash_M)}_{\text{(F1)}} \times \prod_k \underbrace{p\big(\{m_i\}_k \,\big|\, \{z_i\}_k, \theta_k\big)}_{\text{(F2) magnitudes}} \times \prod_k \underbrace{p\big(\{N_{kv}\} \,\big|\, \xi, \theta_k, b_k, f_k, \Lambda\big)}_{\text{(F3) counts}} \\ \times\; \underbrace{p\big(d_{\mathrm{GW}} \,\big|\, \{d_{\mathrm{gal},k}\}, \xi, b_{\mathrm{GW}}, \Lambda, \{\theta_k\}\big)}_{\text{(F4) GW}} \end{aligned} \tag{16}$$

The single structural change is that ξ now appears in (F1), (F3) *and* (F4).

An honest statement about (F2). In the shipped pipeline the magnitude channel is *not* a likelihood factor: the magnitude log-likelihood is defined but referenced nowhere outside its own test, and θ_{sel} enters as an anchored Gaussian prior whose width comes from the offline fit's covariance ($\sigma(\hat{M}_0) = 1.60 \times 10^{-4}$ mag). The shipped hierarchy is therefore empirical Bayes on θ_{sel}, and (16) is the target, not the current object. This matters exactly once: at the unconditional rung of §2.5 the restored shell totals draw θ_{sel} information from the same magnitudes that produced the prior, and the combination is refused.

2.2 (F3): the count likelihood, and the one approximation that buys three things

On solve voxels $v = (p, g)$,

$$N_{kv} \;\sim\; \text{Poisson}\Big[\underbrace{f_{k,p}\, C_k(v;\theta_k)\, \bar{N}_k(v;\Lambda,\theta_k)}_{\bar{\lambda}_{kv}}\; \exp\big(b_k s_v - \tfrac{1}{2} b_k^2 \sigma_v^2\big)\Big], \tag{17}$$

with $\bar{N}_k(v) = \Omega_{\mathrm{pix}} \int_g \bar{n}\, dz$. Two consequences must be stated because both change the completeness side. $f_{k,p}$ appears here *and* in the consumed missing budget $dN_{\mathrm{miss}} = [1 - f_p C]\, \bar{n}\, Q$ — one side only inverts the sign of the inferred incompleteness. And under a per-pixel completeness the base $\bar{\lambda}_{kv}$ is itself a plug-in estimate of the same counts, so (17) would be circular; only bases that are functions of θ rather than of the counts are admissible.

Conditioning on the per-shell totals $T_{kg} = \sum_p N_{kpg}$ factorises the counts exactly,

$$p\big(\{N_{kpg}\} \,\big|\, \xi, \theta\big) = \underbrace{p\big(\{T_{kg}\} \,\big|\, \xi, \theta\big)}_{\text{monopole}} \times \underbrace{p\big(\{N_{kpg}\} \,\big|\, \{T_{kg}\}, \xi, \theta\big)}_{\text{placement}}, \tag{18}$$

and using only the second factor is a deliberate *partial likelihood*: the factorisation is exact, the choice is not free. The discarded monopole is removed to linear order by an explicit projection of the sky-constant coefficients and retains $O(b^2 \mathrm{Var}_{\mathrm{sky}}(s)/2)$ content.

The within-shell linearisation. Write the shell-integrated intensity and its normalised within-shell profile,

$$\Lambda_{pg} = a_g(\theta)\, f_p \int_g \omega_g(z;\theta)\, e^{\,b s_p(z) - \frac{1}{2} b^2 \sigma_p^2(z)} dz, \qquad \omega_g(z;\theta) = \frac{C\bar{n}}{\int_g C\bar{n}\, dz}, \tag{19}$$

with a_g p-independent. Expanding the integral, $\log \int_g \omega_g e^{bs_p} dz = b\langle s_p\rangle_g + \frac{1}{2} b^2 \mathrm{Var}_g(s_p) + O(b^3)$, and *keeping the first term* defines the shell-response operator W to act on the radial *basis rows* rather than around the exponential:

$$\tilde{\phi}_\zeta[g,:] = \sum_n W_{gn}\, \phi_\zeta[n,:], \qquad W_{gn} = \Big(\int_g \mathcal{K}(z|z_n)\, dz\Big)\, \omega_g(z_n;\theta)\, \Delta_n, \tag{20}$$

rows normalised to unity, so that

$$\boxed{\eta_{pg} = b\left(\Phi_{\rm sph}[p] \otimes \tilde{\phi}_\zeta[g]\right)\cdot \xi, \qquad \pi_{pg} = \frac{f_p\, e^{\eta_{pg}}}{\sum_{p'} f_{p'} e^{\eta_{p'g}}}, \qquad \log p_{\rm count} = \sum_g \sum_p N_{pg} \log \pi_{pg}.} \tag{21}$$

This is the load-bearing modelling step, and it is worth being explicit about what it buys and what it costs. *Buys:* η is exactly linear in ξ, so the Kronecker factorisation $\Phi_g = \Phi_{\rm sph} \otimes \tilde{\phi}_\zeta[g]$ is exact, the Hessian below is exact, and the sky-constant subspace is an *exact* gauge direction of (21). *Costs:* the neglected term is precisely $\frac{1}{2} b^2 [\mathrm{Var}_g(s_p) - \langle \sigma_p^2\rangle_g]$, a within-shell variance, which is p-dependent (which is why it does not divide out) and which vanishes identically when s is constant across a shell. *And this is the same condition that makes the completeness parameters cancel* (Prop. 3 below): one approximation controls the separability, the exactness of the Hessian, and the θ-cancellation together, and its controlling scale is the shell width against the photometric-redshift-smoothed radial structure. Moving W onto the basis is also what makes it a genuine forward model: the photometric-redshift kernel is convolved into the *model*, not deconvolved from the data, so the counts remain integer, the multinomial remains exact, and the radial attenuation $\sim e^{-k^2\sigma_\chi^2/2}$ does not bias the tracer bias.

Gradient and Hessian. For (21),

$$\nabla J_\theta = \xi - b \sum_g \Phi_g^\top \big(N_g - T_g \pi_g\big), \tag{22}$$

$$H(\theta) = \mathbb{1} + b^2 \sum_g T_g \Big[\Phi_g^\top \mathrm{diag}(\pi_g) \Phi_g - u_g u_g^\top\Big], \qquad u_g = \Phi_g^\top \pi_g. \tag{23}$$

The rank-one subtraction is not optional: it is positive semidefinite, so omitting it makes H too large, H^{-1} too small, and the Laplace draws *under*-dispersed — exactly the direction that hides the error of Proposition 2. It is separable and therefore free: $u_g = v_g \otimes \tilde{\phi}_\zeta[g]$ with $v_g = \Phi_{\rm sph}^\top \pi_g$. Equation (23) is the *Fisher* information of the multinomial, hence $H \succeq \mathbb{1}$ for any link function; using the observed Hessian instead would forfeit that guarantee under a saturating link, so Fisher scoring is normative rather than convenient.

2.3 (F4): the gravitational-wave host intensity on the same field

Hosts are built from the *same* s with their own bias,

$$\Lambda_{\rm host}(p,z;\Lambda,\theta,\xi) = \psi(z;\Lambda)\, \bar{n}(z;\Lambda,\theta)\, \exp\Big[b_{\rm GW} s(p,z) - \tfrac{1}{2} b_{\rm GW}^2 \sigma^2(p,z)\Big], \tag{24}$$

with ψ the merger-rate weighting. Thinning (24) by the same $f_p C$ gives the catalogued/missing split. The point that a cross-correlation-literate reader checks first is the *catalogued* branch: since hosts trace $e^{b_{\rm GW}s}$ while galaxies trace $e^{b_{\rm gal}s}$, the probability that a specific catalogued galaxy j hosts the event carries an excess-bias factor. The correct prior is

$$p\big(z, p \,\big|\, \mathcal{C}, \xi, \Lambda, \theta\big) \;=\; \frac{1}{\mathcal{Z}(\Lambda,\theta,\xi)}\Bigg[\sum_{j\in\mathcal{C}} \delta_{p,p_j}\,\mathcal{K}_j(z)\,w_j(\Lambda)\,e^{\,(b_{\rm GW}-b_{\rm gal})s(x_j)-\frac{1}{2}(b_{\rm GW}^2-b_{\rm gal}^2)\sigma^2(x_j)} \\ +\;\big[1-f_p\,C(z;\theta)\big]\,\psi(z;\Lambda)\,\bar{n}(z;\Lambda,\theta)\;e^{\,b_{\rm GW}s(p,z)-\frac{1}{2}b_{\rm GW}^2\sigma^2(p,z)}\Bigg]. \quad (25)$$

At $b_{\rm GW} = b_{\rm gal}$ the spike factor is identically unity and the catalogued branch is field-free; away from it, writing the branch field-free is a modelling error that also makes the complete-catalog limit below true by omission rather than by physics. Setting $b_{\rm GW} \equiv b_{\rm gal}$ in a headline analysis is therefore not only a statistics choice; it is the choice that makes Limit I exact.

Structurally (25) is (6) with $Q_p(z)$ replaced by an explicit function of ξ, which is why the change in the evaluation is one contraction: $\log Q \to b_{\rm GW}\,(\text{row\_fac}[p]\cdot\tilde{\phi}_\zeta[n]) - \rho$.

2.4 Double conditioning: the catalog is used once

Theorem 1 (no double use)**.** *Let $\mathcal{C}$ be the catalogued galaxies and $\mathcal{U}$ the missing ones. Then exactly*

$$p\big(\mathcal{C}, d_{\rm GW} \,\big|\, \xi,\Lambda,\theta\big) = \underbrace{p\big(\mathcal{C}\,\big|\,\xi,\theta,\Lambda\big)}_{\text{(F3)}} \times \underbrace{p\big(d_{\rm GW}\,\big|\,\mathcal{C},\xi,\Lambda,\theta\big)}_{\text{(F4)}}, \quad (26)$$

and the second factor may be evaluated with (25) *without any double use of the catalog,* provided the missing intensity is written as a function of ξ alone and never of $\mathcal{C}$.

Proof. (26) is the chain rule. The gravitational-wave factor requires the catalogued hosts — a conditioning event, used as known positions, not re-modelled — and the intensity of $\mathcal{U}$, which by (5) is $[1 - f_p C]\Lambda_{\rm gal}(\xi)$ and involves $\mathcal{C}$ nowhere. The catalog's information about $\mathcal{U}$ has been routed entirely through (F3) into the posterior for ξ. □

Remark 4 (the two ways it breaks, both live). (i) *Empirical-Bayes plug-in.* If the missing intensity is evaluated at a point estimate $\hat{\xi}(\mathcal{C})$ — which is exactly what the current table is — then $\mathcal{C}$ appears inside the gravitational-wave factor, and multiplying by (F3) uses the same counts twice. The present code is safe *only* because it never multiplies by (F3). **Adding the count likelihood without promoting ξ to a latent variable is the single most dangerous intermediate state available, and it must never exist.** (ii) *The monopole, twice.* The calibration that pins $(\log_{10} n_0, \delta)$ is a fit to the *same* counts over $z \in [0.05, 0.3]$, delivered to the run as fixed values; that prior is a surrogate for the first factor of (18). Keeping both it *and* an unconditional Poisson count likelihood squares the monopole information. Exactly two combinations are admissible: conditional counts + external calibration prior, or unconditional counts + a genuinely independent prior on (n_0, δ). The first is recommended; the second carries §2.6's degeneracy.

2.5 Which parameters the field is allowed to couple to

Proposition 3 (exact cancellation). *π_{pg} is exactly independent of θ and Λ if and only if the base is p-separable within the shell — true whenever C is sky-uniform — and the field is shell-constant.*

Otherwise the residual is exactly a within-shell covariance,

$$\frac{\partial \log \pi_{pg}}{\partial \theta} = \frac{\int_g \partial_\theta \omega_g \, e^{b s_p} dz}{\int_g \omega_g \, e^{b s_p} dz} - \sum_{p'} \pi_{p'g} \frac{\int_g \partial_\theta \omega_g \, e^{b s_{p'}} dz}{\int_g \omega_g \, e^{b s_{p'}} dz}, \tag{27}$$

which vanishes identically when $e^{b s_p(z)}$ *is constant across the shell.*

This is the specification of the coupling set, and it is exact. Since $dV_c/dz = (c/H_0)^3 \times \mathrm{shape}(z; \Omega_m)$ and the completeness curve is built h-free, H_0 cancels from (21) identically while $\Omega_m, w_0, w_a, \delta$ and θ_{sel} do not. Three rungs follow, and they are the design space:

rung	structure	parameters reaching the field	per-proposal cost
0	conditional counts, frozen W	none	~ 0
1	conditional counts, live π_{pg}	$\Omega_m, w_0, w_a, \delta, \theta_{\mathrm{sel}}$ (*not* H_0)	$\sim 1\,\mathrm{ms}$
2	unconditional counts	the above, plus H_0 through $n_0(c/H_0)^3$	rung 1 + $\log\det H$

Rung 1 is the target. It satisfies "every cosmological proposal changes the galaxy-field likelihood consistently" for every parameter the galaxy field can honestly constrain, and it is affordable. Rung 0 is its control arm and its fallback: it is exactly rung 1 evaluated at $\Delta\theta = 0$.

2.6 The n_0–H_0 degeneracy: what rung 2 actually costs

Un-conditioning restores the shell-total factor, and the restored channel is

$$T_g(\theta) \;\propto\; n_0 \, (c/H_0)^3 \int_g C_{\mathrm{sel}}(z; \theta_{\mathrm{sel}}) \, (1+z)^\delta \, \mathrm{shape}(z; \Omega_m, w_0, w_a) \, dz, \tag{28}$$

so the *entire* H_0 content of the count channel is the volume–density amplitude $n_0 H_0^{-3}$, exactly degenerate with n_0. It is identified only if n_0 carries an H_0-independent prior — and the shipped calibration does not: it computes n_0 by dividing the summed galaxy weights by a comoving volume integral evaluated at the *fiducial* cosmology. Pinning $\log_{10} n_0$ and then sampling H_0 against an unconditional count likelihood therefore manufactures a standard-density H_0 constraint out of the fiducial cosmology, acting against 22.79M galaxies. The honest sentence, which belongs verbatim in any paper:

> *The galaxy field constrains the shape parameters* $(\Omega_m, w_0, w_a, \delta)$ *and the tracer biases. It constrains* H_0 *only through the volume–density amplitude, which is degenerate with the mean comoving number density; that degeneracy is why* n_0 *is a nuisance and why its prior must be* H_0*-aware rather than a fiducial plug-in.*

2.7 Marginalising the field: the galaxy-side evidence

Proposition 4 (the galaxy-side evidence)**.** *With the whitened prior,*

$$\log \mathcal{L}_{\mathrm{gal}}(\theta) = \log \int d\xi \, \mathcal{N}(\xi; 0, \mathbb{1}_M) \, p(\{N\}|\xi, \theta) \;\simeq\; -J_\theta(\hat{\xi}_\theta) - \tfrac{1}{2} \log\det H(\theta), \tag{29}$$

with no *residual* $(2\pi)^{\pm M/2}$*: the prior normalisation and the Gaussian integral cancel exactly. Equivalently, with* $\ell(\xi, \theta) = \sum_g \sum_p N_{pg} \log \pi_{pg}$*,*

$$\log \mathcal{L}_{\mathrm{gal}}(\theta) = \ell(\hat{\xi}_\theta, \theta) \;-\; \tfrac{1}{2} \big\| \hat{\xi}_\theta \big\|^2 \;-\; \tfrac{1}{2} \log\det H(\theta). \tag{30}$$

The middle term of (30) is an Occam factor and it is *not* removable by the envelope theorem: stationarity gives $\partial_\xi \ell(\hat\xi) = \hat\xi$, not 0, so the envelope theorem removes $\partial\hat\xi/\partial\theta$ from J and not from ℓ. Its size is not academic, and it is the worst case rather than the typical one that happened. Measured on the anchor, $\|\hat\xi\| = 123.5$ at $M = 2520$ (2.46 per mode), and the galaxy-side evidence (30) oscillates by 59.9 nats across θ drawn around the anchor, against a target accuracy of 0.1 nat (§3). Under linear response it is also free:

$$\tfrac{1}{2}\|\hat\xi_\theta\|^2 = \tfrac{1}{2}\|\hat\xi_{\rm ref}\|^2 + \big(S^\top\hat\xi_{\rm ref}\big)\cdot\Delta\theta + \tfrac{1}{2}\Delta\theta^\top\big(S^\top S\big)\Delta\theta, \tag{31}$$

one precomputed vector and one small matrix. Likewise $\log\det H$ is not *assumed* constant: its motion enters multiplied by $M/2 = 1260$ at the shipped rank, so a 0.1% per-mode drift is 1.3 nats, and it is measured directly at the same solves that produce S — the Cholesky factor is already formed. It was measured, and it drifts: 1.47 nats across the prior-drawn θ (§3), so any rung that admits the galaxy-side evidence into a posterior must carry the $\mathrm{tr}(H^{-1}\partial_\theta H)$ term rather than freeze it.

2.8 Linear response: the θ-coupled ensemble

Offline, once. From the implicit function theorem, against the same Cholesky factor,

$$S = \frac{\partial\hat\xi}{\partial\theta} = -H^{-1}\frac{\partial\,\nabla_\xi J}{\partial\theta}\bigg|_{\hat\xi_{\rm ref},\theta_{\rm ref}} \in \mathbb{R}^{M\times n_\theta}, \tag{32}$$

i.e. n_θ extra triangular solves and, in the shipped artefact, 101 kB of storage at $M = 2520$ over five columns $(\hat M_0, \sigma_M, \delta, \Omega_m, b_{\rm gal})$. At $K \geq 2$ each catalog's own selection parameters and its own bias contribute their own *columns* of S: the multitracer generalisation adds terms, it does not restructure anything.

Per proposal. At fixed H, a mean shift is exact:

$$\xi_m(\theta) = \hat\xi_{\rm ref} + S\,\Delta\theta + L_H^{-\top} g_m \tag{33}$$

is an *exact* draw from $\mathcal{N}(\hat\xi_\theta, H^{-1})$ for every θ, with g_m fixed across proposals. There are no importance weights and no effective-sample-size penalty. This is worth stating sharply because the obvious alternative is a trap:

Proposition 5 (the effective-sample-size law for a reused draw set)**.** *If instead one keeps the draws at $\hat\xi_{\rm ref}$ and re-weights, then for a Gaussian mode shift at fixed H,* $\mathrm{Var}[\log w] = \|\hat\xi_\theta - \hat\xi_{\rm ref}\|_H^2$ *exactly, so*

$$\frac{\mathrm{ESS}(\theta)}{M_{\rm draw}} \simeq \exp\Big(-\big\|\hat\xi_\theta - \hat\xi_{\rm ref}\big\|_H^2\Big). \tag{34}$$

The norm is a *sum over all M modes.* A tolerance written per mode as $\|\Delta\hat\xi\|_H/\sqrt{M} < \tau$ implies $\mathrm{ESS}/M \simeq e^{-\tau^2 M}$: at $M = 3780$ and a comfortable-looking $\tau = 0.1$ this is 4×10^{-17}. Reusing a fixed draw set across proposals is inadmissible; the mean-shifted construction (33) is the one to build.

What linear response costs. The correction $S\Delta\theta$ is *member-independent*, and that single fact carries the whole cost model. The member row factors stay static under a 256-fold evaluation concurrency; the only per-proposal additions are one member-independent row-factor contraction (~ 1 ms against the measured 1417 ms baseline, i.e. +0.07%, and a 1.46 MB transient that must be reserved rather than assumed) and a rank-n_θ update of two sky moments. On this catalog/event pairing the correction is also measurably inert on the gravitational-wave channel (5.4×10^{-4} nat, §3), so it ships as insurance against a deeper catalog rather than as a source of signal. It is also *analytically smooth* in θ: no solver, no tolerance, no iteration count, no history — so the discontinuity failure mode of a per-proposal re-solve cannot occur by construction.

Why not simply re-solve. The separable two-stage contraction costs $2G_s N_{\text{pix}} M_{\text{sph}}^2$ for the diagonal part plus $2M_{\text{sph}}^2 G_s M_z^2$ for the radial contraction, and the Cholesky costs $M^3/3$; at $(M_{\text{sph}}, M_z, G_s) = (315, 8, 12)$ the whole anchor build — thirteen Fisher steps plus the moment tables, which dominate — is a measured 41 s on one H100. The solve alone is of order a second, which against the measured 1417 ms baseline is order unity, not the 36× this document first asserted against a baseline 52× too small. The refusal survives the correction, but it has to be argued from the other two costs: a re-solve drags the member row factors into the per-evaluation transient, 3.0 GB at $M_{\text{draw}} = 8$ and 24 GB at $M_{\text{draw}} = 64$ at production concurrency, and it replaces an analytically smooth surrogate by one carrying a solver's tolerance and iteration count. That asymmetry, not wall clock and not elegance, is why Proposition 3 is the load-bearing result.

2.9 The budget normaliser in closed form

The normaliser that enforces the gauge of Proposition 1 against the *consumption* weights is

$$\rho_m(z;\theta,b) = \log \frac{\sum_{p\in\mathcal{F}} \left[1 - f_p C(z;\theta)\right] e^{b f_m(p,z)}}{\sum_{p\in\mathcal{F}} \left[1 - f_p C(z;\theta)\right]} = \log \frac{A_m(z;b) - c\, B_m(z;b)}{P_{\mathcal{F}} - c\, F_{\mathcal{F}}}, \qquad c \equiv C(z;\theta), \quad (35)$$

with $A_m = \sum_{p\in\mathcal{F}} e^{b f_m}$, $B_m = \sum_{p\in\mathcal{F}} f_p e^{b f_m}$, $P_{\mathcal{F}} = |\mathcal{F}|$, $F_{\mathcal{F}} = \sum_{p\in\mathcal{F}} f_p$. The sums run over the *fitted footprint*: off the footprint $f_p = 0$, the consumption weight is unity and $Q \equiv 1$, so that block satisfies the budget identity trivially and must not absorb the footprint's monopole. A_m and B_m are θ-free and are computed once on a grid in b; the only online θ-dependence is the scalar c at each node, so (35) is exact rather than interpolated. This is what removes a full-sky reduction — 25.6 ms in double precision — from the per-proposal path. That reduction was quoted as +50–90% of the entire likelihood; against the measured 1417 ms baseline it is +1.8%, so removing it is worth having and is no longer decisive, and (35) is retained because it is exact rather than because it is fast. Under linear response the moments acquire θ-dependence only through the same projection, $\partial A_m/\partial\theta_j = \sum_p e^{b f_m^{\text{ref}}} (\Phi_p \cdot S_j)$, at 1.1 MB of extra storage.

The identity the design must preserve, exactly, per realisation, is

$$\frac{\sum_{p=1}^{N_{\text{pix}}} \left[1 - f_p C(z;\theta)\right] Q_p(z)}{\sum_{p=1}^{N_{\text{pix}}} \left[1 - f_p C(z;\theta)\right]} \;=\; 1 \qquad \text{for every } z, \text{ member and } \theta. \quad (36)$$

2.10 The member estimator, predicted and then measured

The largest open number in the design was the member spread σ of Proposition 2, because M_{draw} depends on it exponentially. It can be predicted in closed form from objects that already exist. To first order in the field, the gradient of the total event log-likelihood with respect to the latent is

$$\boxed{a \;\equiv\; \nabla_\xi\Big(\sum_i \ell_i\Big) \;=\; b_{\rm GW}\Big(\sum_i \varphi_i\,\Phi_i - N_{\rm obs}\langle\Phi\rangle_{\rm sel}\Big), \qquad \sigma = \big\|L_H^{-1}a\big\|_2} \tag{37}$$

where φ_i is event i's fraction of prior mass in the missing branch and $\langle\Phi\rangle_{\rm sel}$ is the selection-weighted basis average. The norm is the H^{-1} norm, not the Euclidean one: for members drawn as in (10), $\mathrm{Var}_m(\ell_m) = a^\top H^{-1}a$ exactly. The subtraction term is the detection normalisation, and it is what makes (37) a correlation statistic rather than a raw pair count.

The same object a does three jobs, which is the reason to build it first. It predicts σ and hence $M_{\rm draw}$. It gives the closed-form target for the small-bias limit of the estimator: expanding (13) to second order,

$$\operatorname*{LSE}_m \ell_m - \log M_{\rm draw} - \ell(\hat\xi) \;\longrightarrow\; \tfrac12\, a^\top H^{-1}a \;=\; \tfrac12\sigma^2, \tag{38}$$

so prediction and validation become one measurement, checked against truth rather than against a larger draw count. And it supplies the criterion of the next subsection.

The normaliser is not a detail: it corrects both forms at the same order. (37) and (38) are written with the raw basis row Φ_i at the event's voxel. The seam applies the budget normaliser (35) at every evaluation — it is what makes (36) hold — and ρ enters at the same order b^2. Carrying it through the expansion gives, per event at $x_i = (p_i, z_i)$ and with $u_m = \Phi\xi_m$,

$$\log Q_m(x_i) = b_{\rm GW}\big[u_m(x_i) - \langle u_m(\cdot, z_i)\rangle_{\mathcal F}\big] - \tfrac12 b_{\rm GW}^2\,\mathrm{Var}_{p\in\mathcal F}\big(u_m(p, z_i)\big) + O(b_{\rm GW}^3), \tag{39}$$

so *both* orders change: the first contracts the *monopole-projected* row $\psi_i = \Phi(x_i) - \langle\Phi(\cdot, z_i)\rangle_{\mathcal F}$, and the second is a deterministic self-energy. Marginalising,

$$\operatorname*{LSE}_m \ell_m - \log M_{\rm draw} - \ell(\xi{=}0) \;\longrightarrow\; b_{\rm GW}^2 c_\infty + O(b_{\rm GW}^4), \qquad c_\infty = \tfrac12\sum_{ij}\tilde K(x_i, x_j) - \tfrac12\sum_i V(z_i), \tag{40}$$

with $\tilde K(x_i, x_j) = \langle\psi_i, \psi_j\rangle$ and $V(z)$ the footprint variance of the kernel at z. Three things follow. The correction is large: on the pinned fixture $c_\infty = +3.042196$ against the unprojected $+26.931816$, a factor 8.9 and 64 Monte-Carlo standard errors, and the estimator's measured b^2 coefficient lands 1.14 standard errors from c_∞ with the predicted b^4 convergence *rate* verified. Every object in (40) is nevertheless still a pure contraction of the kernel $K = \Phi\Phi^\top$, so *marginalising the Gaussian latent still replaces the field by its kernel* — what dies is the naive pairing, not the structure, and §2.12 is written accordingly. And the reading is physical: the monopole projection removes the field mode the gauge fixing of Proposition 1 conserves, while $-V/2$ cancels the events' self terms $K(x_i, x_i)$, because a common multiplicative normaliser cannot inform an event about its own pixel; at large $|\mathcal F|$, (40) collapses to the strictly off-diagonal $\frac12\sum_{i\neq j}K(x_i, x_j)$.

One further caution about (38) in the form that subtracts $\ell(\hat\xi)$ rather than $\ell(0)$: it is guaranteed positive only if ℓ is linear in ξ, and the shipped seam is not linear at that precision. Measured on the production line it is *negative* at every node below $H_0 = 83.75$ — -5.6×10^{-4} nat at the anchor, where the second-order curvature beats the Jensen term about six to one — and its sign is member-set dependent at $M_{\rm draw} = 8$, which is the statement that eight members do not resolve it. The $b \to 0$ form (40), whose convergence *rate* is verified independently of its level, is the one to test against.

Prediction against measurement. The spread was then measured directly, at $M_{\rm draw} = 256$ under common random numbers, at 34 nodes across the H_0 prior on the production line.

form	σ at the anchor	what it gets wrong
unconditioned Poisson Fisher	6.0×10^{-4}	H; 171× low, and the caveat's sign
Euclidean $\|a\|_2$	1.56×10^{-1}	the norm
(37) with H^{-1} and the real H	1.03×10^{-1}	the monopole projection
measured, $M_{\rm draw} = 256$	$\mathbf{1.49 \times 10^{-2}}$	—

The a vector reproduces between the two implementations to 13 digits, so the entire first-row discrepancy is H: an unconditioned Poisson Fisher was substituted for the shell-conditioned multinomial one, and the caveat attached to it has its sign backwards — a *smaller* H is a larger H^{-1} and therefore a *larger* σ, so the approximation was unsafe in exactly the direction it was claimed to be safe. The two Fishers agree at the top of the spectrum and differ at the bottom, which is where H^{-1} lives: 199 modes below eigenvalue 1.01 against 47. The Euclidean-to-H^{-1} ratio is 1.52, not the 259 that 6.0×10^{-4} implied. The corrected closed form has the sign, the mechanism and the monotone shape right and the level wrong by a median factor 5.5, over-predicting for precisely the reason (40) gives.

Measured, σ is monotone decreasing in H_0: 1.1525 nats at $H_0 = 20$, 1.4882×10^{-2} at the anchor, 1.3814×10^{-3} at 140, with ESS/M from 0.835 to 0.999998. The θ-varying bias of (13) against the 256-member reference is 7.07×10^{-3} nat at the shipped $M_{\rm draw} = 8$ and is met at 0.1 nat by every $M_{\rm draw}$ from 4 to 128 under the antithetic construction, so the marginalisation ships rather than falling back to a fixed realisation. One caveat rides with that and belongs in the same sentence: over 400 random antithetically balanced 8-member subsets of the 256, 22% *fail* the same gate over the full prior, and $M_{\rm draw} = 64$ is the first draw count at which none does. The shipped ensemble passes and under common random numbers it is the only ensemble that exists, but the margin is a property of that artefact rather than of $M_{\rm draw} = 8$; $M_{\rm draw} = 32$ costs +2.25% and buys the difference.

2.11 Two numbers decide the whole design

What decides whether the field must be coupled to the proposal is not whether the fitted field *looks* different — it is whether the *likelihood* is different, in nats. To first order $\Delta \log \mathcal{L} = a \cdot \Delta\hat{\xi}$, and Cauchy–Schwarz gives $|\Delta \log \mathcal{L}| \le \sigma \, \|\Delta\hat{\xi}\|_H$. A per-mode tolerance is therefore indeterminate by $\sqrt{M} = 50.2$ at the shipped rank depending on an alignment between the field displacement (a radial re-weighting of shells) and the event gradient (the localisation volumes) that nothing bounds and that has every reason to be non-negligible, since both live in the same radial direction. Do not bound the alignment; measure it. The two criteria, both in nats, both computable from the same set of solves at prior-drawn θ:

$$\operatorname*{osc}_{\theta} \Delta \log \mathcal{L}_{\rm GW} = \operatorname*{osc}_{\theta} \Big[a \cdot \big(\hat{\xi}_\theta - \hat{\xi}_{\rm ref}\big) \Big], \tag{41}$$

$$\operatorname*{osc}_{\theta} \Delta \log \mathcal{L}_{\rm gal} = \operatorname*{osc}_{\theta} \Big[\ell(\hat{\xi}_\theta, \theta) - \tfrac{1}{2}\|\hat{\xi}_\theta\|^2 - \tfrac{1}{2} \log\det H(\theta) \Big]. \tag{42}$$

Remark 5 (they are not the same number). It is tempting to argue that if the field barely moves then the galaxy-side evidence cannot be informative either. That is false, and the direction of the error matters. The evidence responds to the *full* count residual $r = N - T\pi$ contracted with $\partial_\theta \eta$; the field displacement is that residual *projected onto* span(Φ) and damped by H^{-1}. A θ-direction lying inside span(Φ) is absorbed by the field — large displacement, flat evidence, no constraint. A direction orthogonal to it cannot be absorbed — zero displacement, maximal evidence constraint. Since (27) makes the θ-direction exactly a *within-shell* covariance, and a shell-collapsed basis cannot represent within-shell structure, "small displacement with a live galaxy-side constraint" is the structurally expected case, not an excluded one. (41) and (42) must both be measured and reported.

Both were measured, and they are five orders apart. Over 20 values of θ drawn around the anchor and re-solved with the θ-dependent base inside W, $\mathrm{osc}_\theta\,\Delta\log\mathcal{L}_{\mathrm{GW}} = 5.4\times10^{-4}$ nat — 300 times under the 0.1-nat gate — while $\mathrm{osc}_\theta\,\Delta\log\mathcal{L}_{\mathrm{gal}} = 59.9$ nats. Remark 5 is therefore not a hypothetical but a description of this catalog: the field displacement is invisible to the events and the galaxy-side evidence is enormous. Three consequences, all structural. The θ-coupled rung is a no-op on the gravitational-wave channel here, so it is demoted to an inertness flag and the frozen-anchor ensemble is the deliverable. The linear-response residual is 6.3×10^{-3} nat, so the coupling is available and accurate if a deeper catalog or a larger event set ever moves the first number above the gate — which is why S is still built and stamped. And the log-det drift is 1.47 nats, so a rung that admitted the galaxy-side evidence would have to carry $\mathrm{tr}(H^{-1}\partial_\theta H)$ rather than freeze it. Both oscillations inherit the over-prediction of (37), whose a carries the unprojected Φ_i: 5.4×10^{-4} nat is an upper bound, in the benign direction.

Remark 6 (and the galaxy-side channel must not be quoted as cosmology)*.* $\Delta\log\mathcal{L}_{\mathrm{gal}}$ acts against 22.79M galaxies while the gravitational-wave side carries 259 events, so it can dominate the posterior. But it is a dN/dz-shape constraint, exactly degenerate with the radial evolution index δ, with the population-average photometric-redshift kernel and with the shell binning — all of which are frozen inside W and none of which are sampled. It is therefore reported as a diagnostic and does not enter a headline posterior unless W's own parameters are sampled or profiled. Otherwise the headline result is a galaxy-clustering measurement wearing a dark-siren hat.

2.12 Three limits, one hierarchy

Write the single-event likelihood as

$$\mathcal{L}_i(\Lambda,\theta,\xi) = \int dz\, d\hat{n}\; p_i\big(d_i \,\big|\, d_L(z;\Lambda),\hat{n}\big)\,\frac{\Lambda_{\mathrm{cat}}(\hat{n},z)+\Lambda_{\mathrm{miss}}(\hat{n},z;\xi)}{\int[\Lambda_{\mathrm{cat}}+\Lambda_{\mathrm{miss}}]}. \tag{43}$$

All three established analyses are limits of (43), and marginalising the field is exactly the operation that interpolates between them.

Limit I — complete catalog, good localisation. $f_p C \to 1$ gives $\Lambda_{\mathrm{miss}} \to 0$, and at $b_{\mathrm{GW}} = b_{\mathrm{gal}}$ the catalogued branch of (25) is field-free, so

$$\mathcal{L}_i(\Lambda) \;\longrightarrow\; \sum_{j\in\Omega_{90}^{(i)}} w_j(\Lambda)\, p_i\big(d_i \,\big|\, d_L(z_j;\Lambda),\hat{n}_j\big), \tag{44}$$

the classical weighted-host dark siren, with ξ dropping out of (43) *identically*, member by member. **Exact.** This is a physics identity, not a routing property, and it is the strongest inertness statement the design admits — but only at $b_{\mathrm{GW}} = b_{\mathrm{gal}}$; otherwise the excess-bias factor of (25) survives and the limit is deliberately not an identity.

Limit II — incomplete, good localisation. Both terms live. Because a well-localised event spans $O(1\text{–}10)$ correlation cells at the smoothing scale, the likelihood depends on the field essentially through its point value in the localisation volume, and

$$\log \mathbb{E}_\xi[\mathcal{L}_i] \simeq \log \bar{\mathcal{L}}_i + b_{\mathrm{GW}}\langle s\rangle_i + \tfrac{1}{2}b_{\mathrm{GW}}^2\Big[\mathrm{Var}_\xi\langle s\rangle_i - \langle\sigma^2\rangle_i\Big] + O(b^3). \tag{45}$$

The first-order term is the placement *information*; the second-order term is the placement *uncertainty*, and it is exactly what the member ensemble estimates. This is the regime every real catalog/detector pairing occupies.

Limit III — diffuse localisation: a correlation-function statement. Let the catalogued fraction of each event's prior mass tend to zero while the localisation spans many correlation cells. The galaxy counts are *retained* — they are what carries b_{gal}. Linearising both the event factor and the count factor in the field and marginalising the Gaussian latent jointly,

$$\log\int d\xi\,\mathcal{N}(\xi;0,\mathbb{K})\,e^{(a_{\mathrm{GW}}+a_{\mathrm{gal}})\cdot\xi} = \tfrac{1}{2}\|a_{\mathrm{GW}}\|^2 + a_{\mathrm{GW}}\cdot a_{\mathrm{gal}} + \tfrac{1}{2}\|a_{\mathrm{gal}}\|^2, \tag{46}$$

with $a_{\mathrm{GW}} = b_{\mathrm{GW}}\sum_i \Phi_i$ and $a_{\mathrm{gal}} = b_{\mathrm{gal}}\sum_{p,g}(N_{pg} - T_g\pi_{pg})\Phi_{pg}$. *Marginalising the Gaussian latent replaces the field by its kernel*, and every pair of field-coupled objects contributes $b_a b_b K(x_a, x_b)$: event $\times$ event is the gravitational-wave auto-correlation; event $\times$ galaxy-voxel is

$$b_{\mathrm{GW}}b_{\mathrm{gal}}\sum_i\sum_{p,g}(N_{pg} - T_g\pi_{pg})\,K(x_i, x_{pg}) \;\longrightarrow\; b_{\mathrm{GW}}b_{\mathrm{gal}}\sum_\ell \frac{2\ell+1}{4\pi}\,\mathcal{W}_\ell^i\,\mathcal{W}_\ell^g\,C_\ell^{\mathrm{GW}\times\mathrm{g}}, \tag{47}$$

the harmonic cross-correlation estimator, with the per-event localisation windows $\mathcal{W}_\ell^i$ playing the role of tomographic bin windows and the bias combination $b_{\mathrm{GW}}b_{\mathrm{gal}}$ being exactly what such a measurement constrains; and voxel $\times$ voxel is the galaxy auto-correlation, i.e. the count likelihood itself. **Approximate:** (46)–(47) are the $O(b^2)$ truncation of (43), valid in the Gaussian, weakly clustered regime, and the expansion parameter $b_{\mathrm{GW}}s$ over the events' support is a measured quantity, not an assumed one.

What the budget normaliser does to Limit III, and what survives it. (46) is written with the raw rows Φ_i, and the normaliser (35) corrects it at the same order b^2 (§2.10): every Φ_i is replaced by its monopole-projected ψ_i, and the events' self terms cancel against $-\frac{1}{2}\sum_i V(z_i)$, giving (40). Measured on a pinned fixture the difference is a factor 8.9 and 64 standard errors, so it is a correction and not a refinement. *The unification survives it.* Every term of the corrected form is still a contraction of the same kernel $K = \Phi\Phi^\top$; the event $\times$ galaxy-voxel term is still $b_{\mathrm{GW}}b_{\mathrm{gal}}$ times a kernel sum and still harmonic-transforms to $C_\ell^{\mathrm{GW}\times\mathrm{g}}$; and at large footprint the event $\times$ event term is the strictly off-diagonal $\frac{1}{2}\sum_{i\neq j}K(x_i, x_j)$, which is the same physics stated correctly — a mode that a survey-wide multiplicative normalisation conserves carries no information, and an event cannot inform itself about its own pixel. What does not survive is the naive pairing of a raw kernel against a raw auto-correlation. Any version of this limit written without projecting out the gauge mode over-counts, and by a factor that is measured rather than assumed.

The unification statement, and its honest scope. Limits I–III are not three methods; they are one likelihood at three completeness/localisation regimes, and the field-level hierarchy is the interpolant that is valid at both ends and, unlike either, in the middle. Two scope statements

Algorithm 1 Offline anchor construction (once per catalog set)

Require: catalogs $\{k\}$, areal-completeness maps $\{f_k\}$, anchor $(\theta_{\mathrm{ref}}, \Lambda_{\mathrm{ref}})$, bias grid $\{b\}$

basis: inducing nodes $M_{\mathrm{sph}} \times M_z$; $L \leftarrow \mathrm{chol}(K_{ZZ} + j\mathbb{1})$; form Φ_{sph} and Φ_ζ {never a dense Φ: 47.6 GB at $M = 3780$}
counts: $N_{kpg} \leftarrow \mathrm{histogram}(z_{\mathrm{gal}}; \text{shell edges})$ {integer, exact; edges stamped, never inherited from a global $z_{\max}$}
shell response: W_{gn} by (20), rows normalised; $\tilde{\phi}_\zeta \leftarrow W\phi_\zeta$ {photo-z convolved *forward*, into the basis}
solve: $\hat{\xi}, H \leftarrow$ Fisher scoring on $J_{\theta_{\mathrm{ref}}}$ with (22)–(23) {separable two-stage contraction}
sensitivities: $S \leftarrow -H^{-1}\partial_\theta \nabla_\xi J$ by (32); also $\partial\hat{\xi}/\partial b$ {n_θ+1 triangular solves against the same L_H; 121 kB}
Occam pre-contraction: store $S^\top \hat{\xi}_{\mathrm{ref}}$ and $S^\top S$ {makes (30) free online}
draws: $\xi_m = \hat{\xi} + L_H^{-\top} g_m$, g_m in antithetic pairs, seed stamped
moments: A_m, B_m on the b-grid over $\mathcal{F}$, and their projections $\partial A_m/\partial\theta$, $\partial B_m/\partial\theta$
stamp: content hash over basis, kernel, jitter, N_{side}, f_p, W, shell edges, counts, θ_{ref}, $\{b_k\}$

Algorithm 2 Per-proposal likelihood, rung 1 (linear response)

Require: $(H_0, \Lambda, \{\theta_k\}, \{b_k\}, b_{\mathrm{GW}})$; anchor artefact of Alg. 1

$\Delta\theta \leftarrow \theta - \theta_{\mathrm{ref}}$ {H_0 is absent by Prop. 3}
$\Delta\Xi \leftarrow \mathrm{reshape}(S\,\Delta\theta, (M_{\mathrm{sph}}, M_z))$ {member-independent: this is the whole cost model}
$\log \mathcal{L}_{\mathrm{gal}} \leftarrow \ell_{\mathrm{ref}} + \partial_\theta \ell \cdot \Delta\theta - \frac{1}{2}\|\hat{\xi}_{\mathrm{ref}}\|^2 - (S^\top \hat{\xi}_{\mathrm{ref}})\cdot\Delta\theta - \frac{1}{2}\Delta\theta^\top (S^\top S)\Delta\theta - \frac{1}{2}\log\det H$
$c(z) \leftarrow C(z;\theta)$ {a scalar per redshift node}
$A \leftarrow A^{\mathrm{ref}} + (\partial A/\partial\theta)\Delta\theta$; likewise B; $\rho_m \leftarrow \log[(A_m - cB_m)/(P_{\mathcal{F}} - cF_{\mathcal{F}})]$ by (35)
for each member m **do**
  $\log Q_m(p,z) \leftarrow b_{\mathrm{GW}}\left[(\mathrm{row\_fac}_m[p] + \Delta\Xi[p]) \cdot \tilde{\phi}_\zeta[z]\right] - \rho_m(z)$ {0 exactly for $p \notin \mathcal{F}$ and for $z > z_{\mathrm{depth}}$}
  $\ell_m \leftarrow \textsc{EventTerms}(\log Q_m) - N_{\mathrm{ev}} \log \mu(\log Q_m)$ {same realisation in both, enforced}
end for
$\log \mathcal{L} \leftarrow \log \mathcal{L}_{\mathrm{gal}} + \mathrm{LSE}_m\, \ell_m - \log M_{\mathrm{draw}}$
report member ESS $= \exp(-\sum_m p_m \log p_m)$, $p_m = \mathrm{softmax}_m(\ell_m)$, and the two channels of (41)–(42) separately

ride with this. Limit III is a property of *the hierarchy*; the practical estimator conditions on the galaxy data first, absorbing the counts into $\hat{\xi}$ and H, so the cross term is not a separate object of the estimator. And at rung 0 that conditioning happens at frozen θ, which removes precisely the cosmology dependence of $C_\ell^{\mathrm{GW}\times\mathrm{g}}$ that a cross-correlation measurement uses — which is an argument *for* the θ-coupled rung, not against the unification.

2.13 The algorithm

statement	status	controlling parameter
Thinning independence (5)	**exact**	—
Chain-rule factorisation (26)	**exact**	—
Posterior-mean Q, Lemma 1	**exact**	given the Laplace posterior
Monopole gauge, Prop. 1	**exact**	—
Shell-total factorisation (18)	**exact**	(using one factor is a *choice*)
Kronecker separability and (23) for (21)	**exact**	given the linearisation below
Sky-constant gauge direction of (21)	**exact**	given the linearisation below
θ-cancellation, Prop. 3	**exact iff**	p-separable base *and* shell-constant s
Mean-shifted draws (33)	**exact**	at fixed H
Effective-sample-size law (34)	**exact**	Gaussian mode shift at fixed H
Limit I (44)	**exact**	$f_p C \to 1$ *and* $b_{\mathrm{GW}} = b_{\mathrm{gal}}$
Low-rank $\Phi\Phi^\top \approx K$	approx.	rank vs. $\ell_{\mathrm{ang}}, \ell_\zeta$; hard guard
Within-shell linearisation (21)	approx.	$\frac{1}{2}b^2[\mathrm{Var}_g(s_p) - \langle\sigma_p^2\rangle_g]$
Laplace evidence (29)	approx.	non-Gaussianity of the *count* term on data-rich voxels
Member estimator (13)	approx.	$(e^{\sigma^2} - 1)/2M$, eq. (14); *measured* 7.1×10^{-3} nat at $M_{\mathrm{draw}} = 8$
Deterministic-Q plug-in, Rem. 1	approx.	first order in posterior spread
Renormalise-then-average, Rem. 2	approx.	$\mathrm{Var}_{\mathrm{sky}}(Q)/N_{\mathrm{pix}}$
Frozen H in the draw covariance	approx.	$\frac{1}{2}\sigma^2\epsilon_H$; $< 10^{-5}$ nat at the measured σ
Frozen $\log\det H$	approx.	$\times M/2 = 1260$ per unit relative drift; *measured* 1.47 nat
Linear response $\hat{\xi}_\theta \approx \hat{\xi}_{\mathrm{ref}} + S\Delta\theta$	approx.	second order in $\Delta\theta$
Population-average photo-z kernel in W	approx.	spread of σ_z across the sample
Limit II (45)	approx.	$O(b_{\mathrm{GW}}^3)$
Limit III (47)	approx.	$O(b^2)$ truncation; Gaussianity
$Q \equiv 1$ above z_{depth} and off-footprint	*choice*	zero assumed variance where there is no data
Empirical-Bayes θ_{sel} prior instead of (F2)	*choice*	anchored at 1.6×10^{-4} mag

Table 2: The ledger. The "choice" rows are modelling conventions and are quoted as systematics, never as marginalisations.

2.14 What is exact and what is not

3 What the ladder measured

The ladder of §4 has been run from rung 0 through rung 8. Rungs 0, 3, 4, 5, 5b and the timing half of 6a ran on the production line — 259 events at 4096 posterior samples, 1 067 946 detected injections, the DESI union catalog at $N_{\mathrm{side}} = 64$ (30 470 occupied pixels, 22 787 566 galaxies). The closure campaign, rung 7 and rung 8 ran at mock scale, $N_{\mathrm{side}} = 16$, and are labelled as such. **No production H_0 posterior exists**: that run is held, and every number below is a likelihood-level or mock-level measurement. A reader who takes any of them as "the H_0 result" has read the wrong document.

Can the field move the answer at all? Yes, by two orders of magnitude more than the budget argument allowed. The event-weighted in-support missing-branch mass is $\sum_i \varphi_i = 0.993$ measured through the production likelihood itself — a killer Q table deletes the in-support missing branch from both the event terms and the selection integral, and the event-term delta gives $\sum_i -\log(1-\varphi_i)$ — against a kill threshold of 10^{-3}, with 24.6% of all posterior samples in support. Switching the *existing* table on at the anchor moves $\log\mathcal{L}$ by 10.85 nats across $H_0 \in [20, 140]$ against a threshold of 0.1. Neither leg of the kill criterion fires. The reconciliation is the useful part: the in-support share of the *missing budget* is indeed tiny (1.3×10^{-5} occupied and 3.8×10^{-4} including empty sky), but the in-support share of the *selection integral* is 2.6×10^{-3} and of the event posterior mass 5.7×10^{-3} per event, because both the events and the injections are population-weighted onto exactly the volume the survey covers. The budget share was the wrong statistic, and this is the sentence that replaces it.

A finding about the shipped scans, not about the field. The same measurement establishes that under the production variance guard in its hard form the likelihood is $-\infty$ at *every* H_0 node in $[25, 140]$, in both the structure-on and structure-off arms: the fixed-population 259-event convention delivers a selection N_{eff} of 31–36k against the ~ 92k the GWTC-4/5 criterion demands. The soft-guard wall then contributes $\sim -10^6$ nats and dwarfs the likelihood, so the shipped scans' $H_0 = 139$ rail is guard-shaped rather than likelihood-shaped; on the clean arm the fixed-population posterior peaks at $H_0 = 90 \pm 5$. Every posterior-level comparison in this programme must state which guard convention it ran under, or it is vacuous. The 10.85 nats above is the clean arm.

The count channel, and the gate that changed the deliverable. At the anchor the solve converges to $\mathrm{grad}_\infty = 1.1 \times 10^{-10}$ with $\|\hat{\xi}\|/\sqrt{M} = 2.46$: the clustering is unambiguous. Over 20 values of θ drawn around the anchor and re-solved, the four numbers of §2.11 are the linear-response residual 6.3×10^{-3} nat, the gravitational-wave oscillation (41) 5.4×10^{-4} nat, the log-det drift 1.47 nats and the galaxy-side oscillation (42) 59.9 nats. The first two are the design decision: the θ-coupled rung is inert on the gravitational-wave channel of this pairing, so it is demoted from deliverable to an inertness flag and the frozen-anchor ensemble is what ships, with the quotable systematic "the anchor-frozen field differs from the θ-consistent field by less than 10^{-3} nat of gravitational-wave likelihood content across the selection and budget prior". The last is the dN/dz-shape information the conditioned channel retains against 22.79M galaxies; per D13(a) it stays a diagnostic, and the guard that keeps it there was independently verified to refuse the configuration that would let it in.

The anchor and the seam. The anchor artefact builds in 41 s on one H100 at $M = 2520$ — the plan's gate said "minutes" — into 64 MB carrying $\hat{\xi}$, L_H, the five columns of S, the members, the row factors, the (A, B) moment tables and a content hash; two same-seed builds are bit-identical in every array. On the seam, the budget identity (36) closes to 3.8×10^{-15} at the integral actually consumed, for every z, member and θ, and the survey-global normaliser is member-independent *bit-exactly* at its formation site, which is the gauge fixing of Proposition 1 made literal. The complete-catalog limit (44) is bit-identical per member at $b_{\mathrm{GW}} = b_{\mathrm{gal}}$, off-footprint rows return exactly zero, the legacy goldens are 23/23 bit-exact, 100 repeated evaluations are bit-identical in both arms, and 11 of 11 guard cells refuse or permit exactly what they are supposed to, parsed from the production command line rather than from a fixture.

The member spread. Measured at $M_{\mathrm{draw}} = 256$: 1.4882×10^{-2} nat at the anchor, monotone to 1.1525 at $H_0 = 20$ and 1.3814×10^{-3} at 140; ESS$/M = 0.9998$ at the anchor, 7.998 of 8 members effective. The θ-varying bias at the shipped $M_{\mathrm{draw}} = 8$ is 7.07×10^{-3} nat against a 0.1-nat budget.

§2.10 gives the reconciliation against the two closed forms, both of which were wrong and neither of which is papered over.

Cost. +2.68% of the production baseline for the seam and its eight members; +95.4% for the rung as a whole, of which 97.2% is the per-pixel areal completeness the latent path requires and not the field. §4.2 gives the table.

Two tracers on one field. At mock scale, over 20 realisations with two *disjoint* tracers, one shared ξ and an imposed $b_2/b_1 = 2$: the ratio is recovered in 20 of 20 within 2σ at median 2.00156 and $\sigma_r = 0.00836$, against a deliberately decoupled two-field variant that returns median 1.456 at $\sigma_r = 1.07984$ — 129 times looser and biased at the same time, with a cross-tracer correlation of 0.999944 shared against exactly 0 decoupled. So the shared field is what makes the bias ratio measurable at all, and the seam that could not be built on the table path (§1.7) is closed by deleting the producer. The overlap arm is the warning that rides with it: 5% shared membership already breaks the recovery at 10σ, and the quoted σ_r *shrinks* rather than growing, because the estimator reads duplicated galaxies as extra independent information. An overlapping pair is not merely less accurate, it is confidently wrong — which is why D9's disjointness is a modelling requirement and not a convenience.

Above the survey depth. Delivered as a sensitivity scan and never as a posterior, at mock scale: over assumed amplitudes $amp(z > z_{\rm depth}) \in \{0, 0.05, 0.1, 0.2, 0.4\}$ the H_0 median moves by less than $0.01\,{\rm km\,s^{-1}Mpc^{-1}}$ and the 90% width by less than 0.06, monotonically and linearly in the assumed amplitude, with the growth-shaped profile indistinguishable from the step at fixed amplitude. For scale, turning the *fitted* field on at all moves the same median by 0.94 and the width by 0.71 on the same realisation. The assumption about 99.4% of the missing budget is worth about 1% of what the fitted part is worth, which is the bounded systematic D7 asks for.

The closure that failed, and what it actually measures. This is the part a proposal is tempted to leave out. Of the mock closure's four tiers, two pass and two fail. Tier A — fitted against true field, gauge-fixed — gives slope 0.99908 ± 0.00433 at Pearson $r = 0.99945$, with $\|\hat{\xi} - \xi_{\rm true}\|/\sqrt{M}$ matching $\sqrt{{\rm tr}H^{-1}/M}$ to 2.4%: the estimator recovers the field it is given. Tier D — perturb f_p by its own dispersion, add a 5% fibre-collision proxy, double the angular lengthscale, give the field a non-Gaussian tail — moves H_0 by at most $+0.390\sigma$ against a 0.5σ tolerance. But Tier B puts $H_0^{\rm true}$ outside the 90% interval in *every* arm, and Tier C rejects uniformity of the posterior quantiles at $p = 0.0111$ with 9 of 24 realisations outside the 99% band, an overconfidence ratio of 2.55.

The failure does not localise where it would have to for the field to be the cause, and three independent measurements say so. First, the control arm with *no* field and no completion at all is overconfident by 2.35: whatever is wrong is wrong before the latent path is reached. Second, splitting each mock's random stream immediately before the event draw gives a law-of-total-variance decomposition in which 92% (no field) and 82% (field) of the excess variance is already present with the catalog, the field, f_p, the mask, the depth map and the anchor all held byte-identical — the excess is the *event draw at fixed catalog*, and the decomposition reproduces Tier C's own ratios from an independent design. Third, the diagnosis that the member draws were under-dispersed was implemented rather than argued: the $b_{\rm gal}$ dispersion of the plan's ${\rm Cov}(\xi) = H^{-1} + s_b^2 vv^\top$ was added, Tier C re-run at $n = 50$, and the overconfidence moved from 2.5515 to 2.5526 — +0.04%, in the wrong direction — with the 90% width ratio at 1.00065 and the inflated arm wider in 24 of 50. Sweeping s_b over three decades refutes the diagnosis outright: inflating the member spread by 4.54 makes the interval 0.6% *narrower*, so the response has the wrong sign and no value of s_b closes the tier. The mock's own synthetic parameter-estimation widths are calibrated to 6%, which cannot supply a factor 2.4 either, and the $N_{\rm obs}$ lever arm gives spread $\propto N^{-0.31\pm0.16}$ where per-event over-sharpness would predict 0.5 and a per-realisation common mode 0 — so the mechanism inside the event channel is not yet separated.

The honest statement, and the one this document makes, is therefore: *the code is validated and the mock's parameter-estimation calibration is not.* The identities close, the estimator recovers the field, the guards refuse what they should, the systematics are bounded — and the coverage gate that would license a cosmology claim is not met, for a reason measurably upstream of everything proposed here. Nothing in this programme should be quoted as validated by a closure that failed, and the production run must not be read as blessed by one.

What is still unmeasured. The production H_0 posterior, and with it the two posterior-level criteria K2 and K8. The mock's event-channel defect, which needs the ~ 4 GPU-hour design that separates per-event over-sharpness from a per-realisation common mode. A second member seed — the resampling that puts 22% of 8-member ensembles outside the marginalisation gate draws from within one artefact's 256 members and is not an independent check of that artefact. And a derivative kink at $H_0 = 98.83$ in a fixture-scale sweep, present identically in the table path and in the latent path and absent when the completion channel is switched off, whose mechanism was not identified.

4 Implementation

4.1 The ladder

Legacy behaviour is bit-identical throughout, enforced by a golden test at 10^{-12} relative tolerance in exact mode; regenerating that golden is forbidden at every step. Two early exits are built in by design, at day 3 and at day 14.

step	scope	gate	days
0	**Can the field move the answer at all?** $\sum_i \varphi_i$ and σ from (37); the oscillation of $\log \mathcal{L}$ across the H_0 prior with the *existing* table switched on at the anchor; a three-column cost baseline (no structure / table / latent); the paper-by-paper novelty search	**kill criterion K0**; all four artefacts committed. Nothing downstream is usable until this lands	3
1	Basis module: Kronecker design, named jitter convention, shell response (20), sky moments, prior variances; builders rewired to call it	Kronecker identity to 10^{-13} against a factored reference at production rank; builder outputs byte-identical	3

step	scope	gate	days
2	Areal completeness as data: f_p at $N_{\mathrm{side}} = 64$; counts as an inference input; $C_p = f_p C(z)$ on *both* sides behind its own flag and its own golden	counts round-trip exactly; sky area to 1%; flag off is bit-identical; flag on moves the missing budget in the predicted direction and by the predicted magnitude; **K8** on a non-railing configuration	5
3	**Count channel, θ-shaped from the start**: (21), (22)–(23), Fisher-scoring solve, sensitivities (32) built from the stacked K-tracer objective, antithetic draws	separable Hessian vs. dense to 10^{-10}; solver vs. dense Newton to 10^{-8}; within-shell Jensen residual reported; **both criteria** (41)–(42), the log-det drift, and the linear-response residual — **kill criterion K9**	5
4	Anchor artefact: $\hat{\xi}$, L_H, S, Occam pre-contractions, (A, B) and their projections, members, row factors, stamp	build completes at $M = 3780$ in minutes; moment tables to 10^{-6}; projected moments to the same scale; draws reproduce the posterior mean; hash reproducible	4
5	The evaluation seam, inert by default: latent leaves built and barriered outside the traced body; closed-form ρ; memory accounting on both the static and the reserved-transient path	every golden cell bit-identical with the flag off; **complete-catalog limit** (44) **bit-identical per member at** $b_{\mathrm{GW}} = b_{\mathrm{gal}}$; off-footprint rows return exactly zero; reservation within 10% of measured peak	4
5b	Member spread: predict σ by (37), then measure it at $M_{\mathrm{draw}} = 256$ across the H_0 prior	predicted vs. measured σ reconciled; **closed-form limit** (38); θ-varying bias < 0.1 nat at the chosen M_{draw}; **K5**	2
6a	**The compressed ensemble** (control arm, and the fallback): rung 0 live at the anchor; budget identity (36); bias-parameter inversion; provenance replaced by the artefact stamp	budget identity to 10^{-12} for all z, members and θ; closure tiers A–D; overhead inside the accuracy budget; **K2**, **K4**	5
6b	**The θ-coupled likelihood** — the deliverable: linear response (33), projected moments, the evidence (30), the two-channel report	at $\Delta\theta = 0$, bit-identical to 6a; linear-response residual rechecked at production rank; no adjacent-θ discontinuity over 10^4 points; measured overhead $\sim$ 1 ms	5
6c	*Optional*: unconditional counts, restoring H_0 through $n_0(c/H_0)^3$; Laplace log-det live; saturating link	agrees with 6b where conditioning is benign; **K10**; gradient-norm stopping rule with the residual inside the oscillation budget	5

step	scope	gate	days
7	Multitracer $K \geq 2$: per-tracer bias, completeness, counts and columns of S; the realisation identifier retires	$K = 1$ bit-identical; bias ratio recovered within 2σ over 20 realisations; **shared-field coupling demonstrably tighter than two independent fits**, shown against a deliberately decoupled variant	4
8	Support and amplitude profile above z_{depth}	delivered as a *sensitivity scan* — H_0 shift versus assumed amplitude — never as a marginalised posterior	3
10	Performance and the production campaign	chunk plans; sample compaction below z_{depth}; the run	3
11	The paper	the day-3 novelty search is cited; the scope sentences of §1 and the H_0 sentence of §2.6 appear verbatim	3

Critical path as planned: 31 days to the compressed ensemble, 36 to the θ-coupled likelihood, with exits at day 3 and day 14. What happened is that the day-14 exit fired on its benign branch, which changed the deliverable rather than stopping the work: rungs 0 through 8 have run, rung 6b is an inertness flag, and rungs 10 and 11 are open. Rung by rung:

rung	what it returned
0	$\sum_i \varphi_i = 0.993$ and $\mathrm{osc}_{H_0} = 10.85$ nats: **K0 does not fire**. Cost baseline measured, and the shipped scans shown to be guard-shaped. Novelty search over ~ 22 papers: the shared-multitracer differentiator not found, the budget-preserving one anticipated only in disjoint pieces. Its own $\sigma = 6.0 \times 10^{-4}$ was later shown wrong by 171× (rung 5b) and its 3027 ms baseline to be an A100-80 number, not H100 (rung 6a).
1–2	basis module and f_p as data. f_p alone costs +92.8% of the baseline — the single largest cost in the programme, and no field is involved.
3	the count channel, θ-shaped: residual 6.3×10^{-3}, (41) 5.4×10^{-4} nat, log-det drift 1.47 nat, (42) 59.9 nat. **K9 fires on its benign branch**: 6b becomes an inertness flag and **6a is the deliverable**.
4	the anchor artefact at $M = 2520$: 41 s per build on one H100, 64 MB, bit-identical across same-seed rebuilds, $\mathrm{grad}_\infty = 1.1 \times 10^{-10}$.
5	the seam, inert by default: goldens 23/23 bit-exact, (44) bit-identical per member at $b_{\mathrm{GW}} = b_{\mathrm{gal}}$, off-footprint rows exactly zero.
5b	$\sigma = 1.4882 \times 10^{-2}$ nat at the anchor and ≤ 1.1525 anywhere; bias 7.07×10^{-3} nat at $M_{\mathrm{draw}} = 8$. **K5 unfired**; the marginalisation ships and the fixed-realisation fallback is not triggered.
6a	budget identity to 3.8×10^{-15} and member-independent bit-exactly; seam +2.68%; determinism 100/100 bit-identical; guards 11/11; Tier A and Tier D **pass**, Tiers B and C **fail** for a reason upstream of the field (§3).
6b	demoted by rung 3 to the $\Delta\theta = 0$ inertness flag.
7	$K = 2$ on one shared ξ: b_2/b_1 recovered 20/20 within 2σ at median 2.00156, 129× tighter than two independent fits; 5% tracer overlap breaks it at 10σ.
8	the $amp(z > z_{\mathrm{depth}})$ sensitivity scan: H_0 median moves < 0.01 and the 90% width < 0.06 across assumed amplitudes 0–0.4.
10–11	not run; the production 259-event posterior is held.

4.2 Cost

The baseline is measured, and the first thing to say about it is that the number this document was written against was wrong twice over. The 27.5–49.3 ms figure belongs to a different configuration; the production 259-event line costs 1417 ms per likelihood call with no structure at all, on one H100, value-only path, 256 concurrent evaluations, 10.4 GiB of 72.7 GiB already static. An intermediate figure of 3027 ms was carried for several rungs and is an A100-80 number rather than an H100 one; percentages built on 27.5 ms deflate by 52×, not 110×. Everything below is measured on that line, at $M_{\mathrm{draw}} = 8$, under the shipped guards:

arm	structure	f_p	ms/eval	vs. baseline
production today	none	no	1417.5	—
areal completeness only	none	yes	2732.2	+92.8%
table mode, 8 members marginalised	Q table	no	1428.7	+0.8%
latent, 8 members marginalised	latent	yes	2770.2	+95.4%

Two numbers come out of that table and they say opposite-sounding things, so both are stated. *The seam is nearly free.* At equal f_p treatment the latent field plus its eight-member ensemble is +37.95 ms: +1.39% of the f_p baseline and +2.68% of the no-structure baseline, and members cost 4.92 ms each thereafter — +2.25% at $M_{\mathrm{draw}} = 32$, +4.5% at 64, against the latent likelihood one actually runs. *The configuration is not free, and the cost is not the field.* Turning the rung on against the shipped baseline roughly doubles the likelihood, and 1314.8 of the 1352.7 ms — 97.2%

— is $C_p = f_p C(z)$, which the latent path requires and which any run adopting it would pay with no field at all. That row appears in no cost table written before the ladder ran, and it is the number an operator budgeting the production run needs.

Three structural facts hold the rest together. The compression from a member *table* cube to member *row factors* is a factor $N_{\mathrm{grid}}/M_z = 136$ at the shipped $M_z = 8$ — measured on the anchor artefact, 1.06 GB against 7.8 MB at $M_{\mathrm{draw}} = 8$ over the 30 470 fitted pixels — which is what makes a defensible M_{draw} affordable at all under (14). The full-sky budget reduction leaves the per-proposal path entirely, in closed form, by (35). And the θ-correction is member-independent, so the resident latent arrays stay static (+140 to +225 MB) and the only per-evaluation transient is the 1.46 MB row expansion of $S\Delta\theta$, i.e. ~ 375 MB at 256-fold concurrency — reserved explicitly, not assumed. Under any per-proposal re-solve variant the second and third fail simultaneously: the moments regain θ-dependence, the row factors become transient (3–24 GB), and the evaluation goes memory-bound before it goes compute-bound. That, and not wall clock, is the argument against a re-solve.

4.3 When to stop

These are the criteria as they were written before the ladder ran; the outcomes in bold are what §3 measured against them. Two of them — K2 and K8 — are posterior-level and remain unscored, because the production run is held and because the mock that would score them fails its own coverage tiers for a reason upstream of the field.

K0 (day 3).

If the event-weighted in-support prior mass is below 10^{-3} *and* switching the existing table on at the anchor moves the log-likelihood by less than 0.1 nat across the H_0 prior, the field cannot move the posterior. Stop. The deliverable is the bounded systematic, and it is three days away rather than thirty-six. **Did not fire.** The budget-side context was right and irrelevant: the in-support missing budget is 1.3×10^{-5} of the total and only 24.6% of event posterior samples fall in support at all — but the event-weighted mass is $\sum_i \varphi_i = 0.993$ and the oscillation is 10.85 nats, each two orders past its threshold, because the events sit where the catalog is. Replacing the budget share by the event-weighted share is the single most useful thing this rung returned.

K2.

If neither latent-on versus latent-off nor θ-coupled versus frozen moves the H_0 median by 0.1σ or the 90% interval width by 5%, on closure *and* on a non-railing production configuration, stop at that rung and publish the bound as a systematic. **Unscored**: the production run is held, and the closure's coverage tiers fail in the no-field arm as well, so no H_0 comparison on that mock is currently readable.

K3.

Fitted-versus-truth slope below 0.8 at the guard-compliant rank: the basis is collapsing (the pathological value is 0.04). **Did not fire**: measured 0.99908 ± 0.00433 at Pearson $r = 0.99945$.

K4.

Cannot fire on cost — the seam is +2.68% of the measured baseline, an order of magnitude under any budget written against it. Had it fired: added wall beyond the accepted budget with the accuracy target unmet, descend one rung at a time — θ-coupled $\to$ frozen ensemble $\to$ fixed-realisation systematic — rather than jumping to the weakest fallback.

K5.

Member effective sample size below 2 of M_{draw} with the θ-varying bias unmeetable at $M_{\text{draw}} \leq 128$: do not ship it as a marginalisation. **Did not fire**: 7.998 of 8 effective members at the anchor and 7.07×10^{-3} nat of bias, met at every M_{draw} from 4 to 128. It ships as a marginalisation.

K8.

The areal-completeness change is *expected* to be large — the 18 682 empty pixels alone move the all-sky missing budget by $\sim +45\%$ — so if it moves the H_0 posterior by more than 1σ with the field off, that change is a result in its own right and ships as its own arm; the field programme continues behind it. **Unscored at the posterior level**, but its cost is now known and is the largest in the programme: $+92.8\%$ of the baseline, 34.7 times the entire latent seam.

K9 (day 14).

If (41) exceeds 0.1 nat and the linear-response residual cannot be brought below 0.1 nat with a second-order term and multiple anchors, then a re-solve is required and a re-solve is not affordable: ship the frozen ensemble and quote the measured θ-dependence as a systematic. Symmetrically, if (41) is below 0.1 nat the coupling adds nothing on the gravitational-wave side — but by Remark 5 that says nothing about (42), which is gated separately. **The benign branch fired**: 5.4×10^{-4} nat against 59.9 nats on the galaxy side, exactly the asymmetry the remark predicted, and it is what demotes rung 6b.

K10.

Does not arise: the unconditional rung was not taken. If it is taken and n_0 is not sampled under an H_0-aware prior (or reparameterised H_0-invariantly), and likewise for θ_{sel} against its anchored magnitude prior, any H_0 shift is an artefact of the fiducial calibration. Refuse the arm rather than publish it.

4.4 What not to attempt

A full per-proposal Newton re-solve — of order the entire likelihood on the measured baseline rather than 36 times it, so the refusal rests on the memory (it makes the member arrays transient at 3–24 GB) and on the loss of an analytically smooth surrogate, not on wall clock. Reusing one draw set across proposals with importance weights ($\text{ESS}/M = 4 \times 10^{-17}$ at a per-mode displacement that reads as comfortably tight). Adding the count likelihood while leaving the field a plug-in $\hat{\xi}(\mathcal{C})$ — Remark 4(i). Materialising Φ (47.6 GB at $M = 3780$). Sampling ξ in the outer chain (the production run has 20 free dimensions; $M \geq 2520$ is not a perturbation on that). Widening shells past the photometric-redshift scatter. And building the multitracer table producer: the field-level path makes it unnecessary, and it would inherit that producer's missing budget gauge and its 1.34 Gpc rank floor. One caution on rank: a feasibility argument made at $M \sim 10^2$ is made at a rank the resolution and isotropy guards forbid — feasibility here comes from the correction being member-independent and of rank n_θ, not from a small M.

5 Decisions required

D1. Which rung? *(a)* conditional counts, frozen response: no θ coupling; the control arm and the fallback. *(b)* conditional counts, linear response: coupling for $\Omega_m, w_0, w_a, \delta, \theta_{\text{sel}}$, no H_0 — was recommended conditional on the day-14 criterion. **The criterion returned benign** (5.4×10^{-4} nat, §3), so on this catalog/event pairing (a) *is* (b) to within 10^{-3} nat and the recommendation becomes (a) with (b)'s machinery built, stamped and carried as insurance against a deeper catalog. *(c)* unconditional counts: adds H_0 only through $n_0(c/H_0)^3$, admissible only with D6 resolved and §2.6 stated in the paper. *(d)* unconditional counts with a per-proposal re-solve: infeasible. *Trade:* conditioning gives up the dN/dz constraint on the budget, restored explicitly as a prior rather than as a by-product of the field.

D2. Jitter convention for the Kronecker basis. *Recommend* an absolute, amplitude-independent 10^{-6} on both factors, stamped. *Cost:* latent-mode Q differs from any legacy table by $\sim 2\times10^{-3}$ at the guard rank, so every migration comparison references a same-convention rebuild.

D3. Smoothing scale. *Recommend* the isotropic $\sim 190\,\mathrm{Mpc}$ kernel ($\ell_{\mathrm{ang}} = 0.2$, $\ell_\zeta = 0.039$, $M_{\mathrm{sph}} = 315$, $M_z = 8$–12). *Cost:* the analysis is explicitly a 190 Mpc-smoothed density field. *Alternative:* keep 50 Mpc radially and accept a 4:1 anisotropic kernel measuring 20% of its own radial signal — which must then be stated as such.

D4. Per-pixel areal completeness. *Recommend* f_p on *both* sides with $C_p = f_pC(z)$ in the missing budget — exact for areal masking. *Fallback:* restrict the count channel to a mask-homogeneous sub-footprint, exact without touching the completeness side, losing $\sim 30\%$ of area. Not an option: ignoring it, which inverts the sign of the inferred incompleteness.

D5. Marginalisation-accuracy budget, hence M_{draw}. *Recommend* 0.1 nat of θ-varying bias across the H_0 prior, with M_{draw} set against (14) and σ measured in the H^{-1} norm of (37). *Measured:* $\sigma = 1.49 \times 10^{-2}$ nat at the anchor and ≤ 1.15 anywhere, bias 7.07×10^{-3} nat at $M_{\mathrm{draw}} = 8$, so the budget is met and the fixed-realisation alternative is not triggered. *Cost, measured:* the seam plus eight members is +2.68% of the baseline and members are 4.92 ms each thereafter — +2.25% at $M_{\mathrm{draw}} = 32$, +4.5% at 64, not the +31/125/250% projected against a baseline 52× too small. *Recommend* $M_{\mathrm{draw}} = 32$ rather than the shipped 8 for one reason: 8 passes on the ensemble that will actually run, but 22% of alternative balanced 8-member ensembles do not, so the margin is a property of one artefact. Thirty-two costs two percent and buys the difference.

D6. Budget uncertainty. *Recommend* sampling $(\log_{10} n_0, \delta)$ under the calibration covariance, systematics-inflated, rather than the current plug-in from the same counts. **Required, not recommended, under D1(c)**: the calibration must be re-derived as a function of H_0, or n_0 reparameterised to an H_0-invariant combination such as n_0h^{-3}.

D7. Above the survey depth. *Recommend* $Q \equiv 1$ with zero variance in the headline analysis, with the amplitude profile delivered as a sensitivity scan quoted as "H_0 shift under an assumed amplitude", never as a marginalised posterior — because no measurement in this programme constrains the field there. *Measured* (§3): across assumed amplitudes 0–0.4 the H_0 median moves by < 0.01 and the 90% width by < 0.06, about 1% of what turning the fitted field on at all is worth on the same realisation. The systematic is bounded and small.

D8. Is the gravitational-wave bias free, and what does that imply for the catalogued branch? *Recommend* $b_{\mathrm{GW}} \equiv b_{\mathrm{gal}}$ in the headline run: it makes Limit I exact and the complete-catalog test a genuine physics identity, at zero cost — and that test now runs, bit-identical per member (§3). Wherever b_{GW} is free — the mock campaign, the multitracer tier — the excess-bias factor $e^{(b_{\mathrm{GW}}-b_{\mathrm{gal}})s(x_j)}$ **must** be written on the catalogued spike weights. Writing that branch field-free is a modelling error, and it is the first thing a cross-correlation-literate referee checks.

D9. Tracer overlap at $K \geq 2$. *Recommend* a disjoint partition. The event-side mixture is safe under overlap; the count-side product is not — and the size of the failure is now measured (§3): 5% shared membership breaks the bias-ratio recovery at 10σ while the quoted σ_r *shrinks*, because duplicated galaxies read as extra independent information. Disjointness must be established as a property of the catalogs, not assumed, and any future rung admitting overlap has to model the shared membership rather than tolerate it.

D10. Field family for the headline run, and whether to keep the table path. The production table is radial, and the radial builder is a per-pixel quasi-Newton loop that cannot move inside the likelihood at any rung — so adopting the latent path *is* a change of family, by necessity rather than preference. *Recommend* running four arms of one comparison before choosing — radial table / low-rank table / frozen latent / θ-coupled latent — and keeping the table path indefinitely as the golden-pinned reference. Note that the $K \geq 2$ parametric-selection *table* arm cannot be built today at any K (§1.7).

D11. The novelty search. Resolved by scheduling: it was the opening item of work, not the closing one, and it is done — ~ 22 papers individually assessed on 2026-08-13, recorded in `pr0/NOVELTY_SEARCH.md`, which is the artefact any priority sentence in this document must be read against. The shared multi-tracer completion realisation is not found in the literature; the budget-preserving marginalisation is anticipated only in disjoint pieces (deterministic mean-one renormalisation; realisation marginalisation through the event terms of a complete catalog; analytic free-monopole marginalisation), and the conjunction — budget-constrained stochastic realisations through *both* the event terms and the selection normalisation — is not. The headline claim therefore survives only with its qualifying clauses; an unqualified "first field-level dark sirens" would not.

D12. What is the headline? Strip what is already scored as not novel and the budget-preserving marginalisation reduces to a gauge fixing (Prop. 1) plus a correctness requirement (same realisation in the event terms and the normalisation, already enforced); with the mandatory scope caveats — $Q \equiv 1$ with zero assumed variance off the footprint and above z_{depth}, so the marginalisation is over the volume the survey sees and not over the budget — what remains is *"we marginalise over the angular placement of missing hosts inside the surveyed volume at fixed budget."* *Recommend (a):* headline the **unification** — Limits I–III as one likelihood, with the field as the object that makes the interpolation real — and demote the budget-preserving marginalisation to the enabling clause it is. *(b)* headline the marginalisation and accept that it may read as a methods note. *(c)* defer until the day-3 search returns. The search has returned (D11) and it supports (a), with two amendments the measurements force: the interpolant is made real by the field itself rather than by its θ-coupling, which is measurably inert here; and Limit III must be stated in its monopole-projected form (§2.12), which is a factor 8.9 from the naive one.

D13. Does the galaxy-side evidence enter the headline posterior? It acts against 22.79M galaxies while the events number 259, and it is a dN/dz-shape constraint degenerate with δ, with the photometric-redshift kernel and with the shell binning — all frozen inside W. *Recommend (a):* report it as a diagnostic; it does not enter the headline posterior unless W's own parameters are sampled or profiled. *(b)* admit it and sample/profile those parameters — real work, and the only honest route to quoting cosmology from it. *(c)* admit it with W frozen: refused.

D14. Is the magnitude channel a likelihood factor or an anchored prior? Today it is an anchored Gaussian prior at 1.6×10^{-4} mag; the magnitude log-likelihood exists but is never evaluated. *Recommend (a):* keep empirical Bayes, state it plainly, and let the rung-dependent guards carry the consequences — zero work, honest. *(b)* implement it as a likelihood factor and drop the anchored prior: this closes (16) exactly and is what would permit the unconditional rung without double-counting θ_{sel}, at the cost of real work.

The two decisions that gate everything else were D1 and D12, and both have been answered. D1: the day-14 criterion is benign, so the deliverable is the frozen-anchor ensemble with the θ-coupled machinery built and inert. D12: the search supports headlining the unification, in the corrected form. What now gates the programme is neither of them — it is the coverage failure of §3, which is a defect in the mock's event channel rather than in anything proposed here, and which must be closed before any H_0 number from this line is quoted as validated.